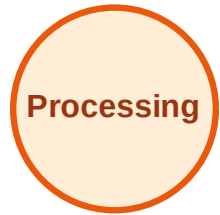


BFS Traversal

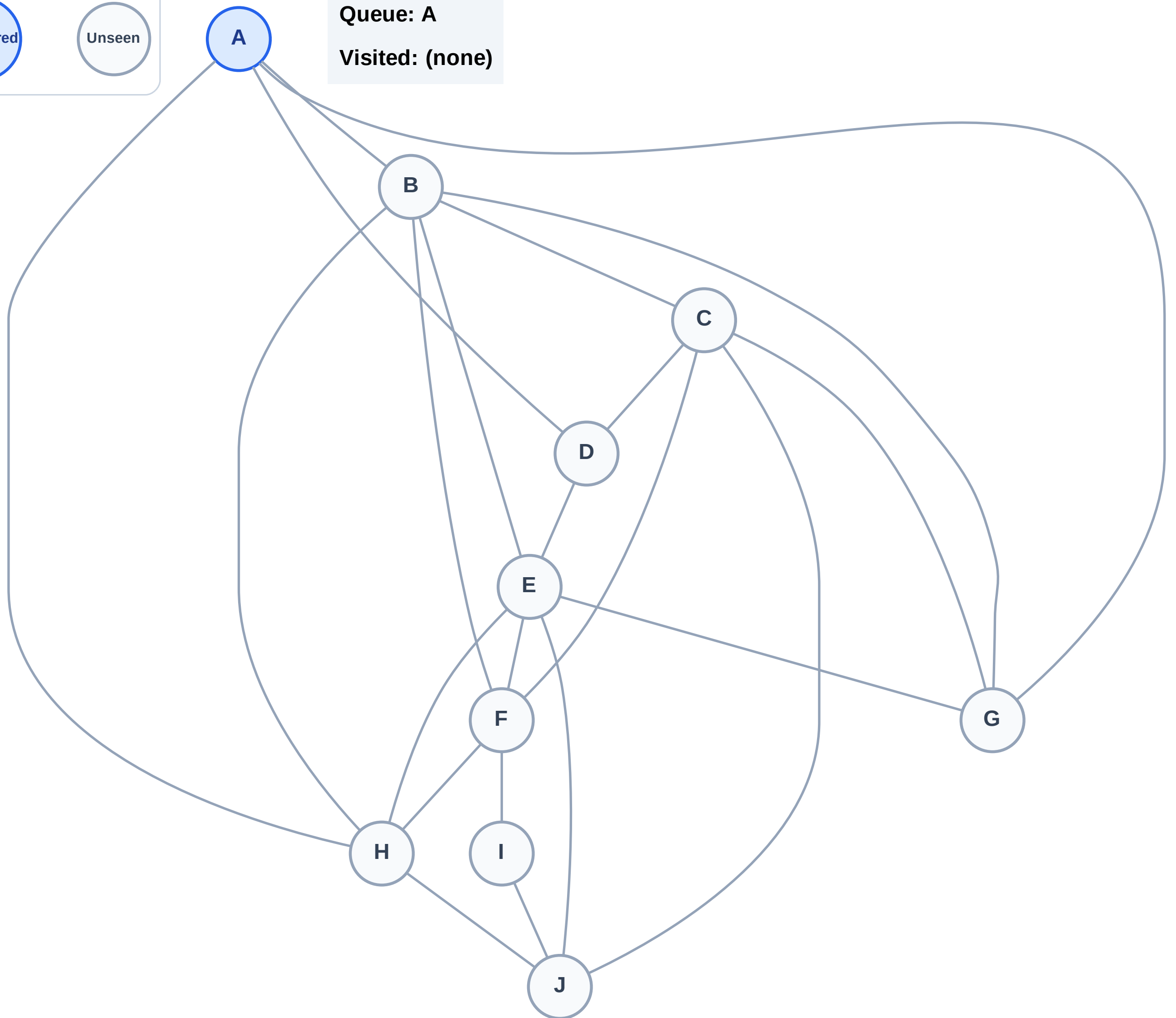
Step 1: Start at A

Legend



Queue: A

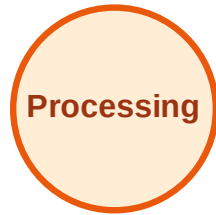
Visited: (none)



BFS Traversal

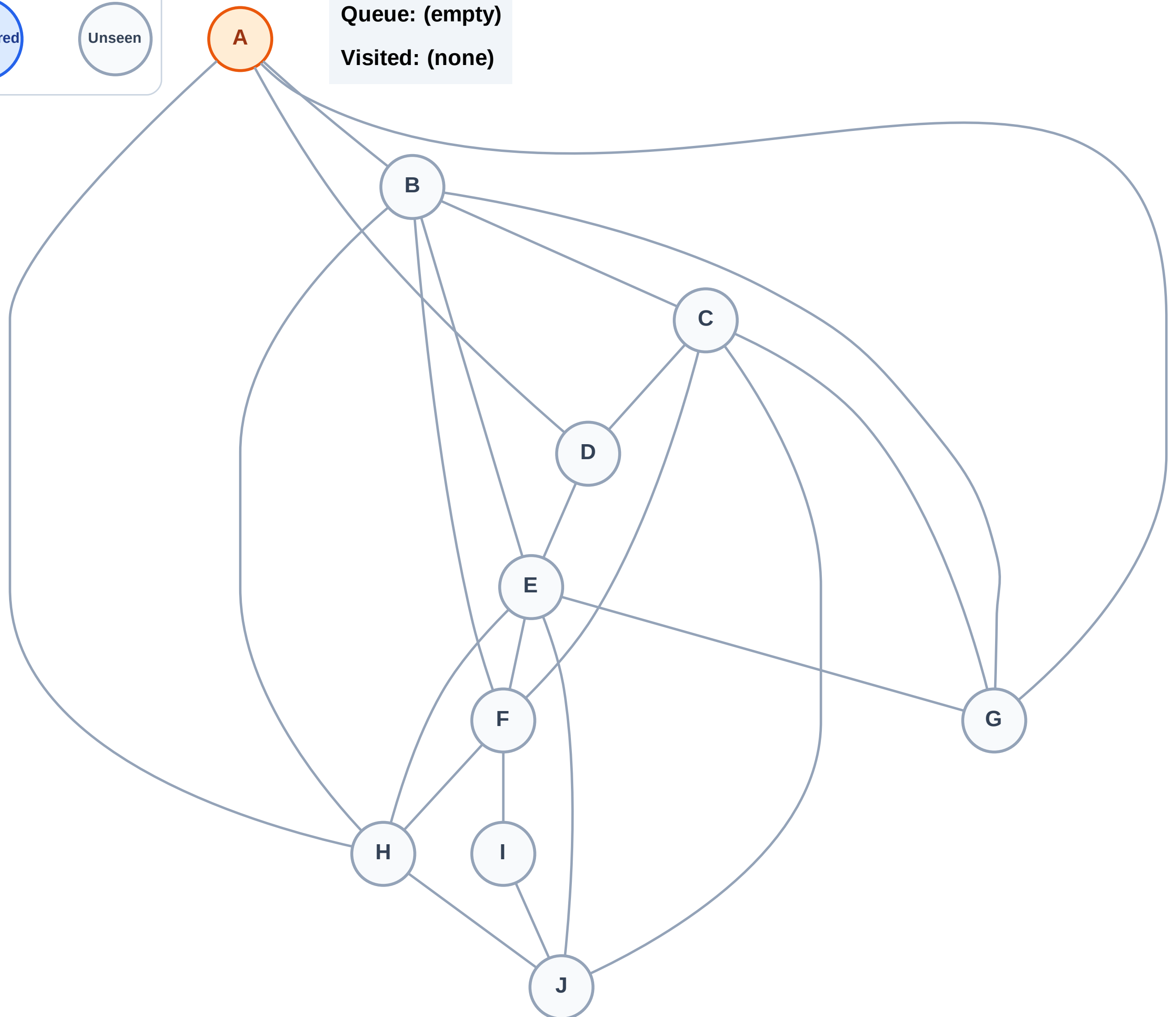
Step 2: Process node A

Legend



Queue: (empty)

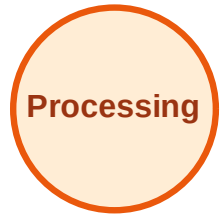
Visited: (none)



BFS Traversal

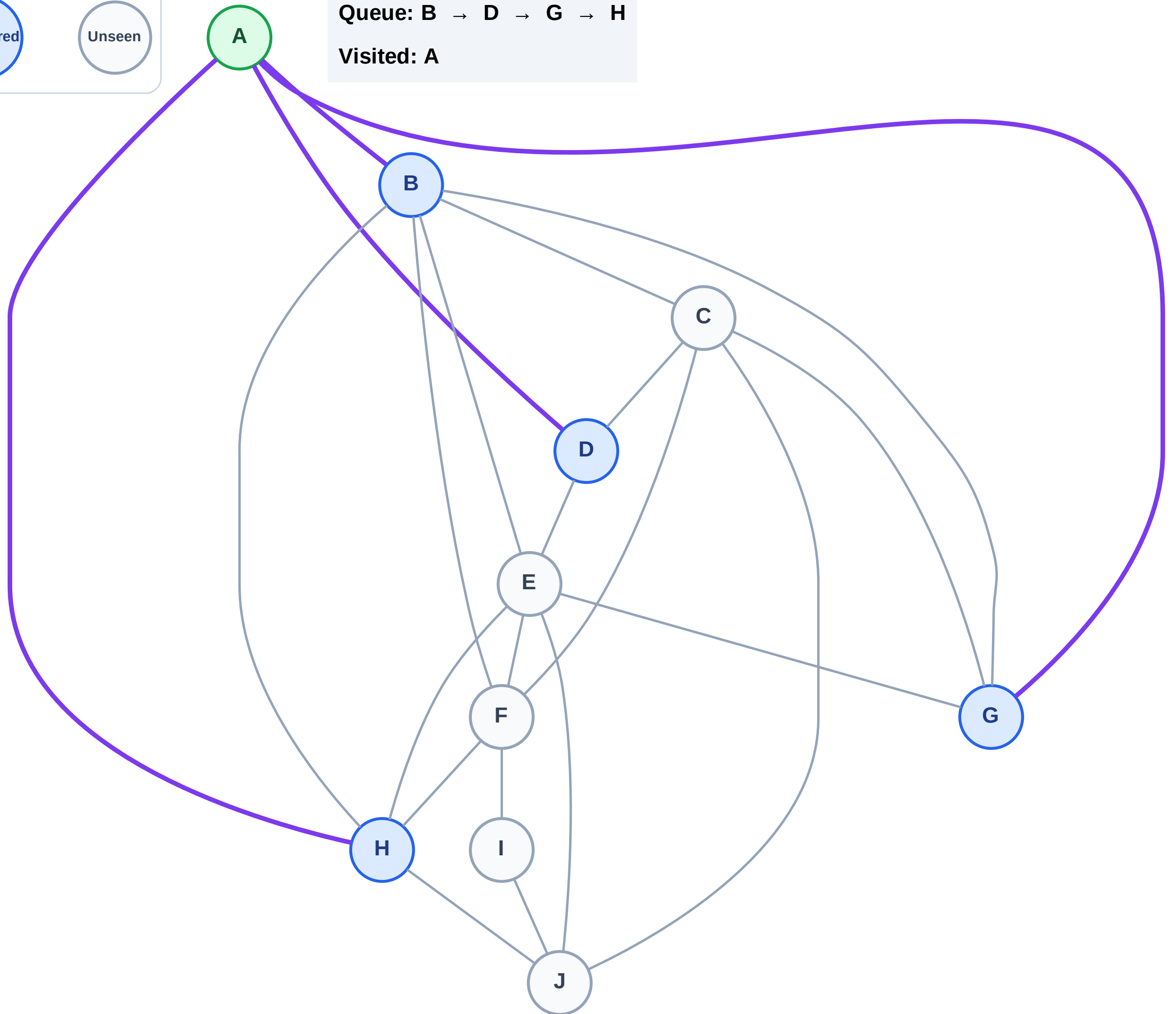
Step 3: Discover B, D, G, H from A

Legend



Queue: B → D → G → H

Visited: A



BFS Traversal

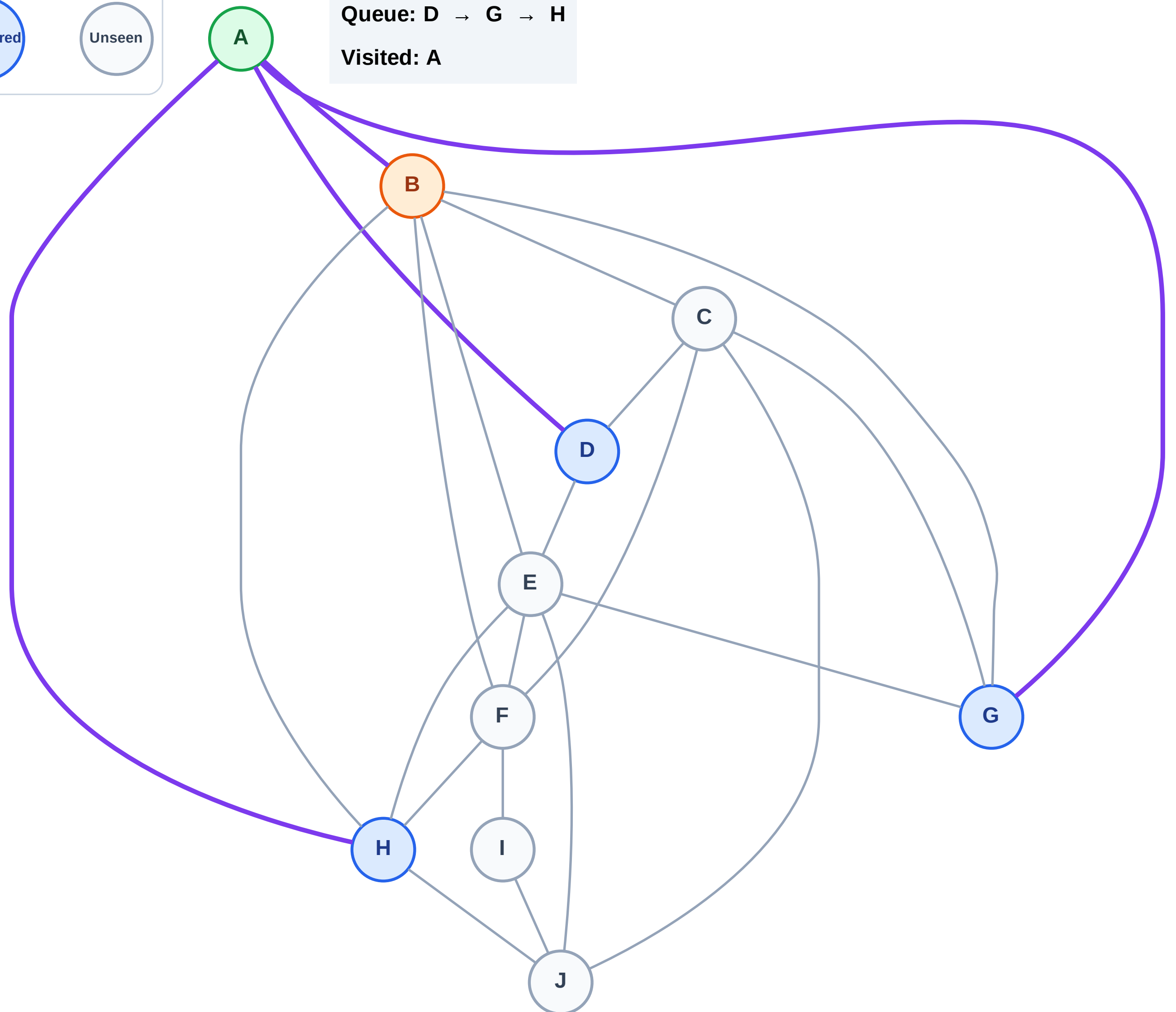
Step 4: Process node B

Legend



Queue: D → G → H

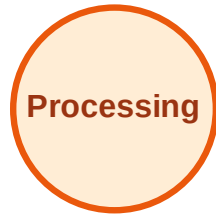
Visited: A



BFS Traversal

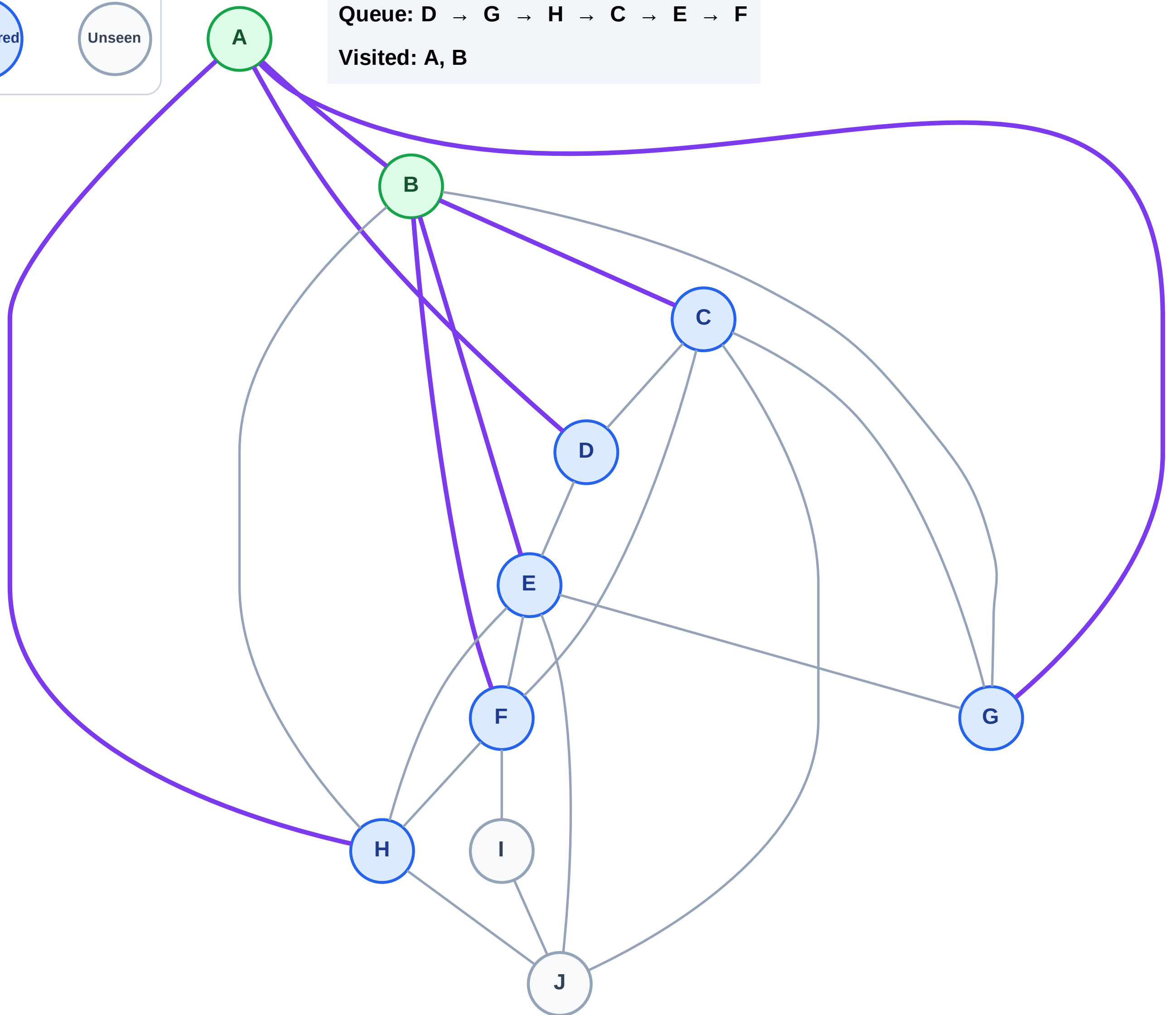
Step 5: Discover C, E, F from B

Legend



Queue: D → G → H → C → E → F

Visited: A, B



BFS Traversal

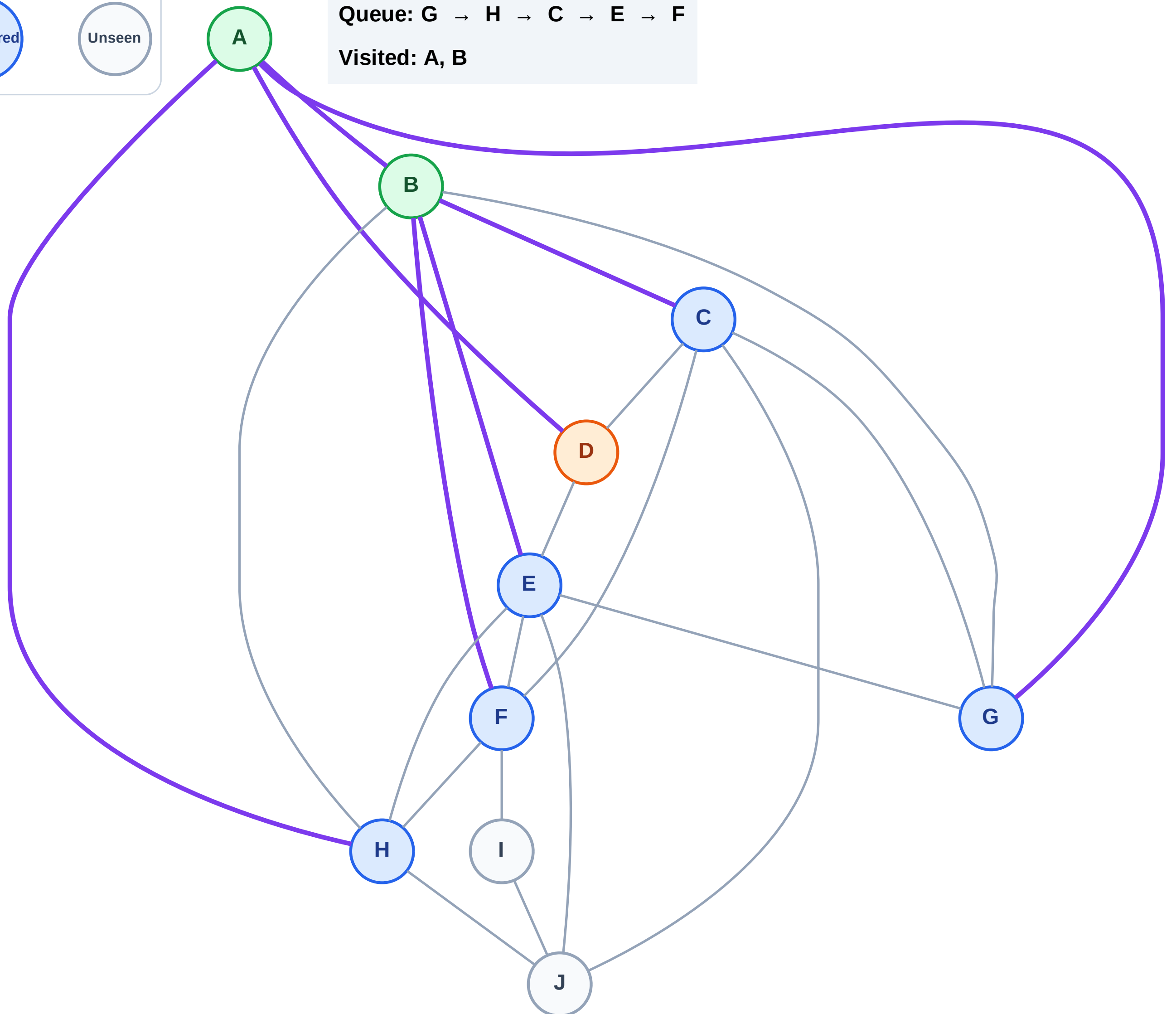
Step 6: Process node D

Legend



Queue: G → H → C → E → F

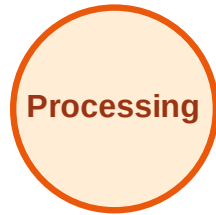
Visited: A, B



BFS Traversal

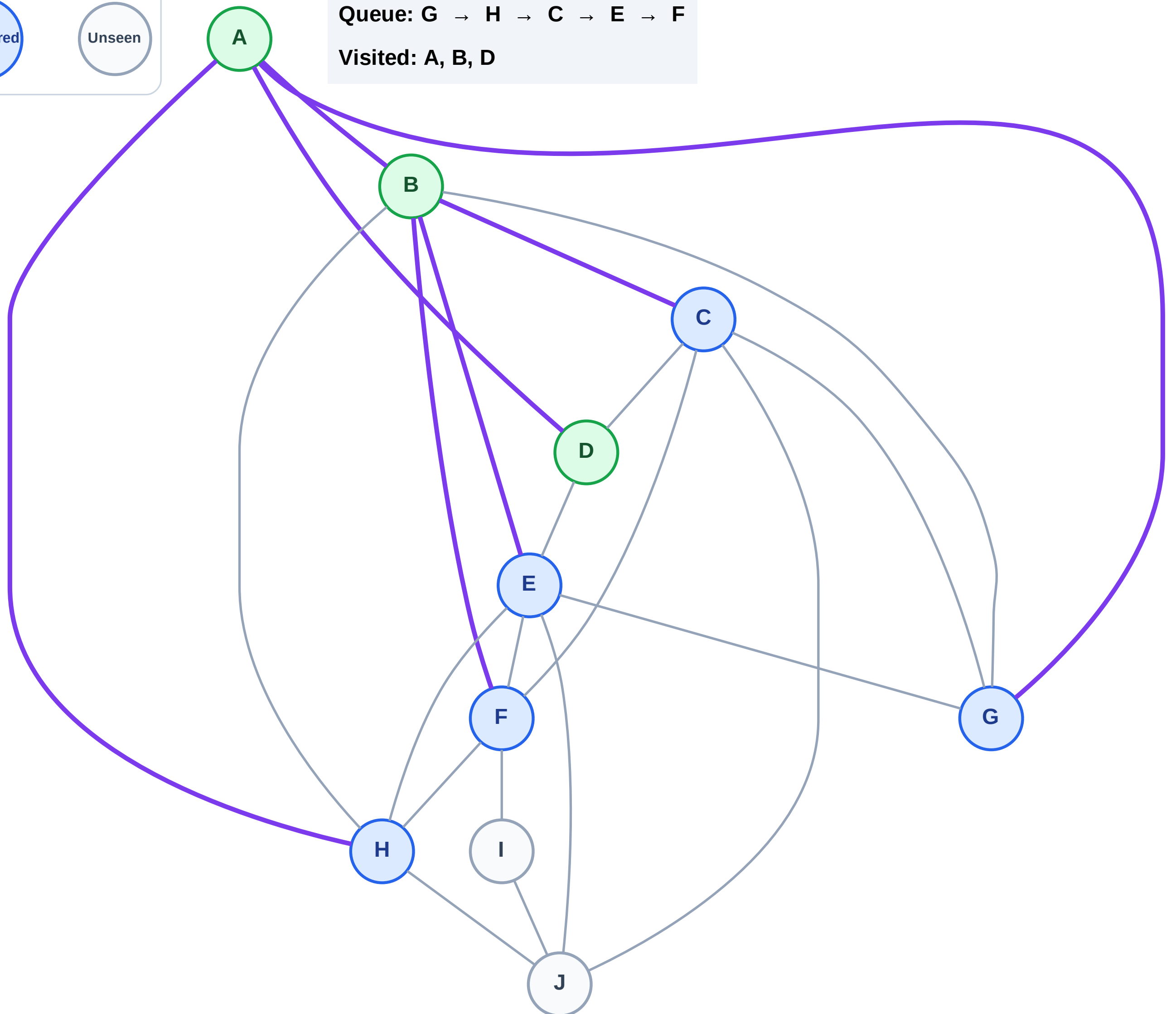
Step 7: No new neighbors from D

Legend



Queue: G → H → C → E → F

Visited: A, B, D



BFS Traversal

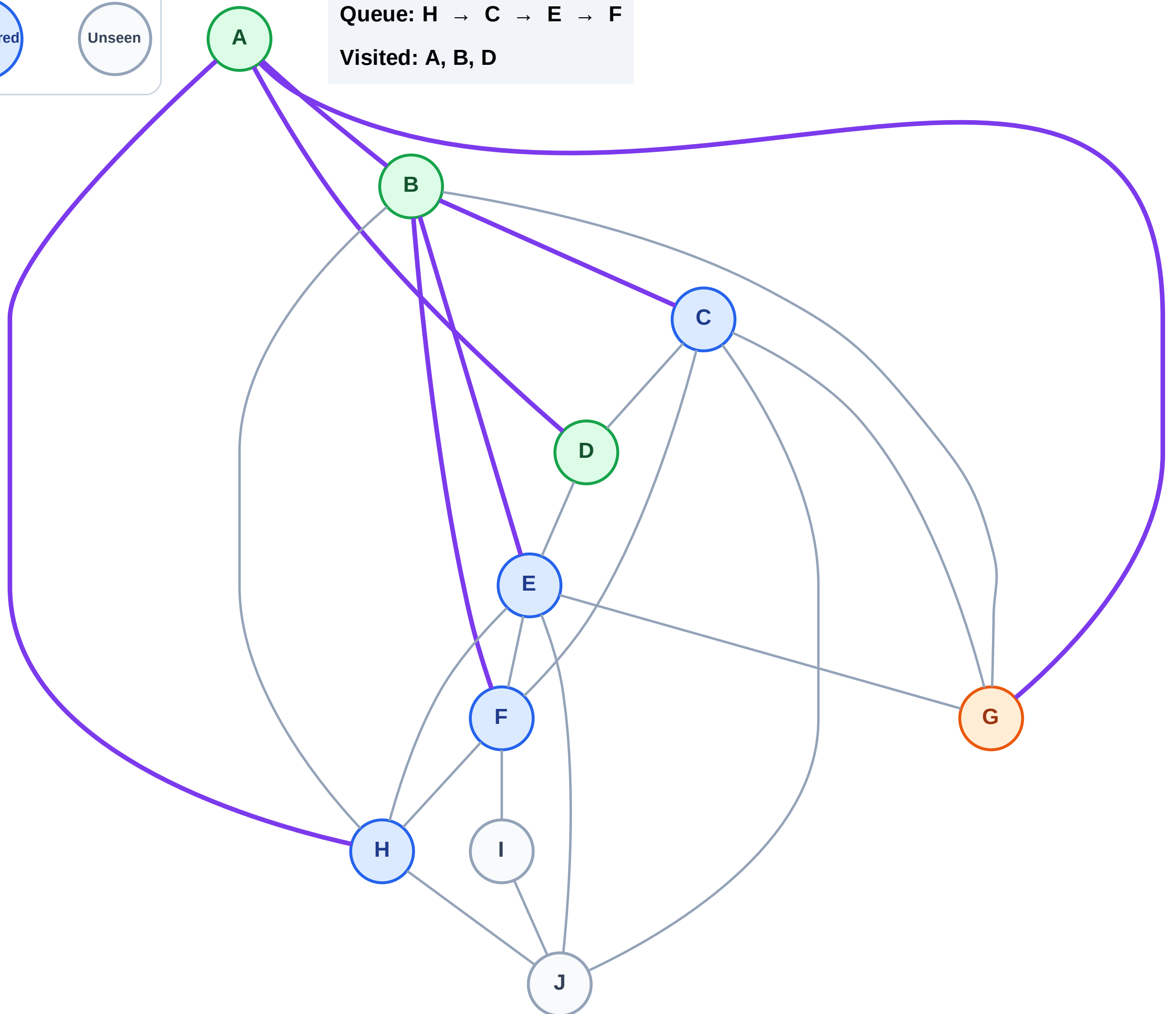
Step 8: Process node G

Legend



Queue: H → C → E → F

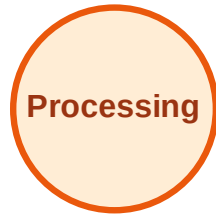
Visited: A, B, D



BFS Traversal

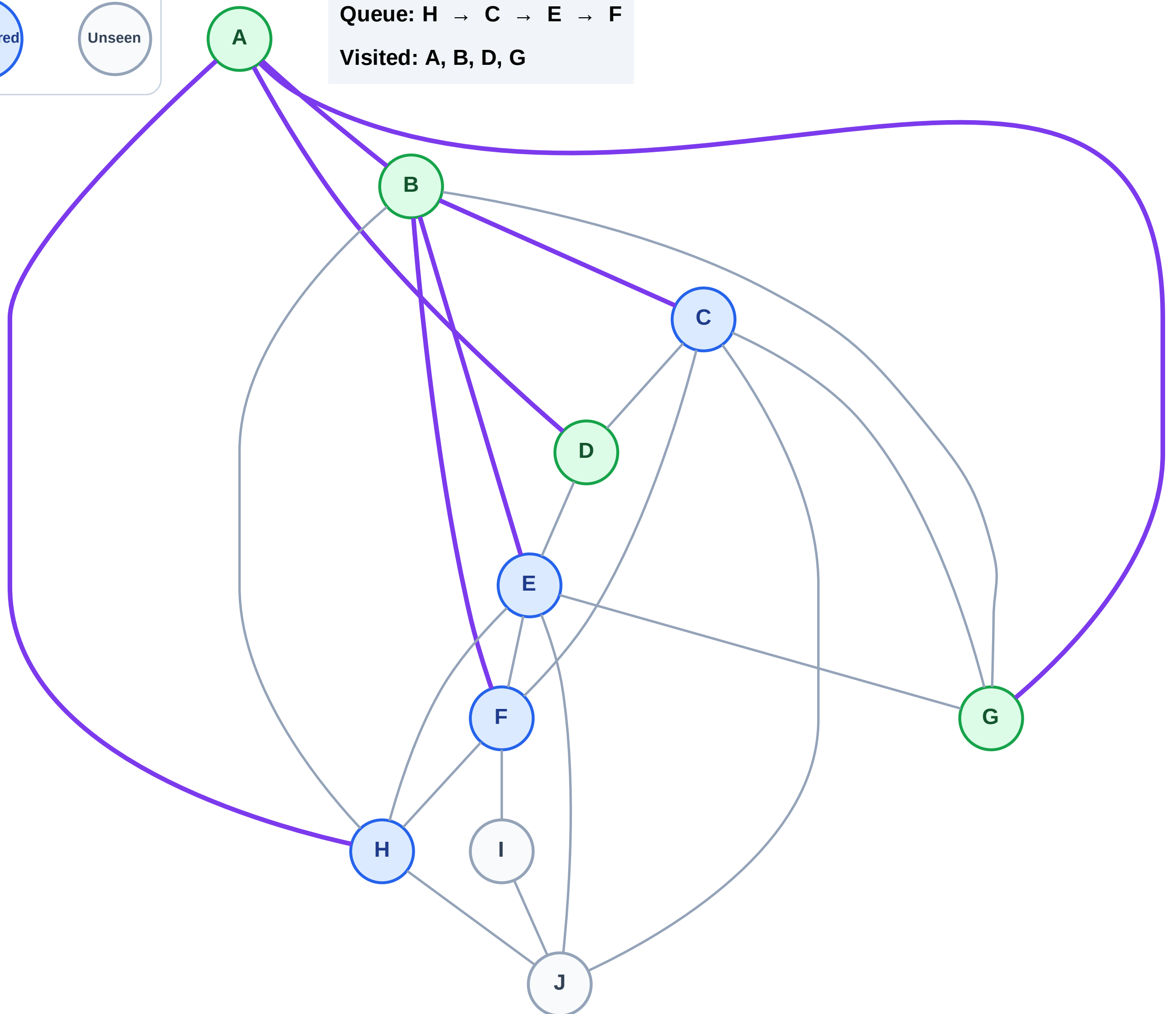
Step 9: No new neighbors from G

Legend



Queue: H → C → E → F

Visited: A, B, D, G



BFS Traversal

Step 10: Process node H

Legend

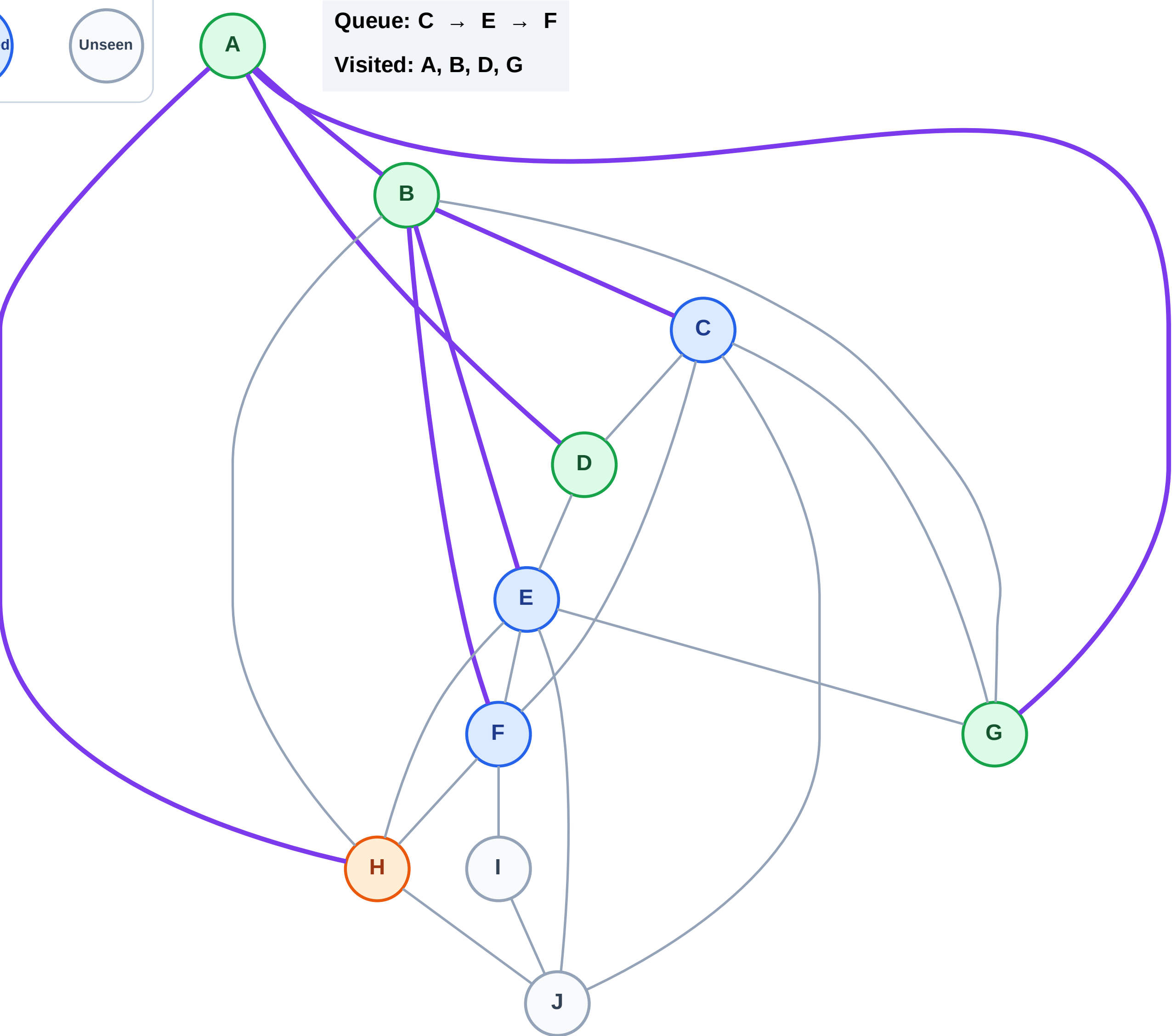
Complete

Processing

Discovered

Unseen

Queue: C → E → F
Visited: A, B, D, G



BFS Traversal

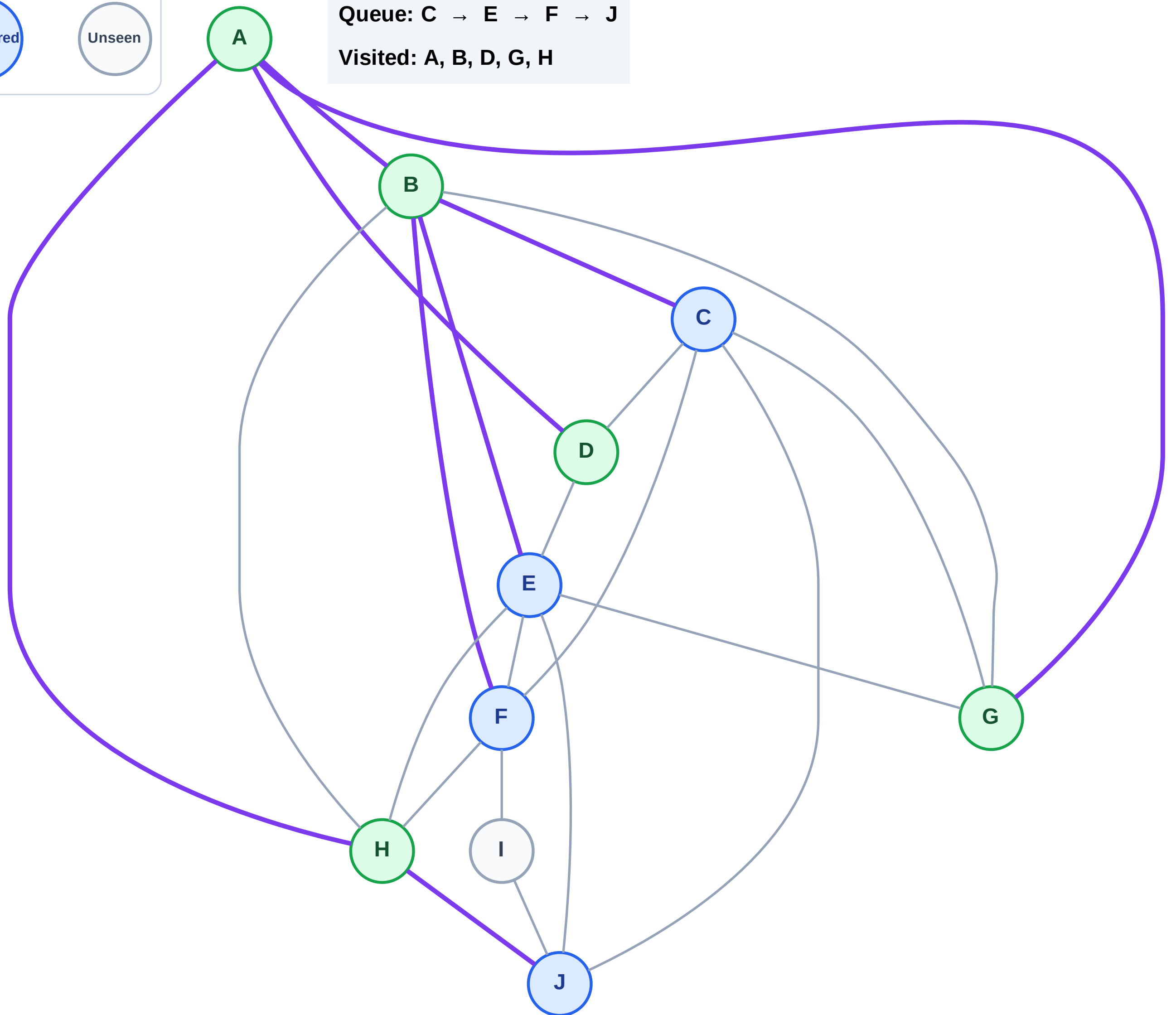
Step 11: Discover J from H

Legend



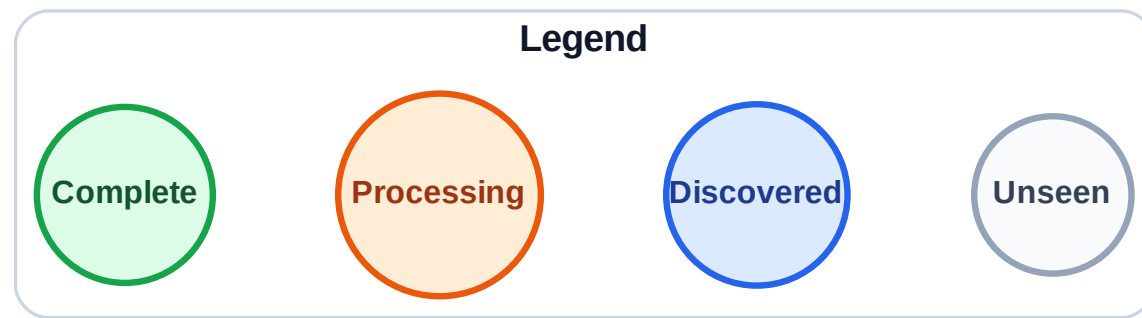
Queue: C → E → F → J

Visited: A, B, D, G, H

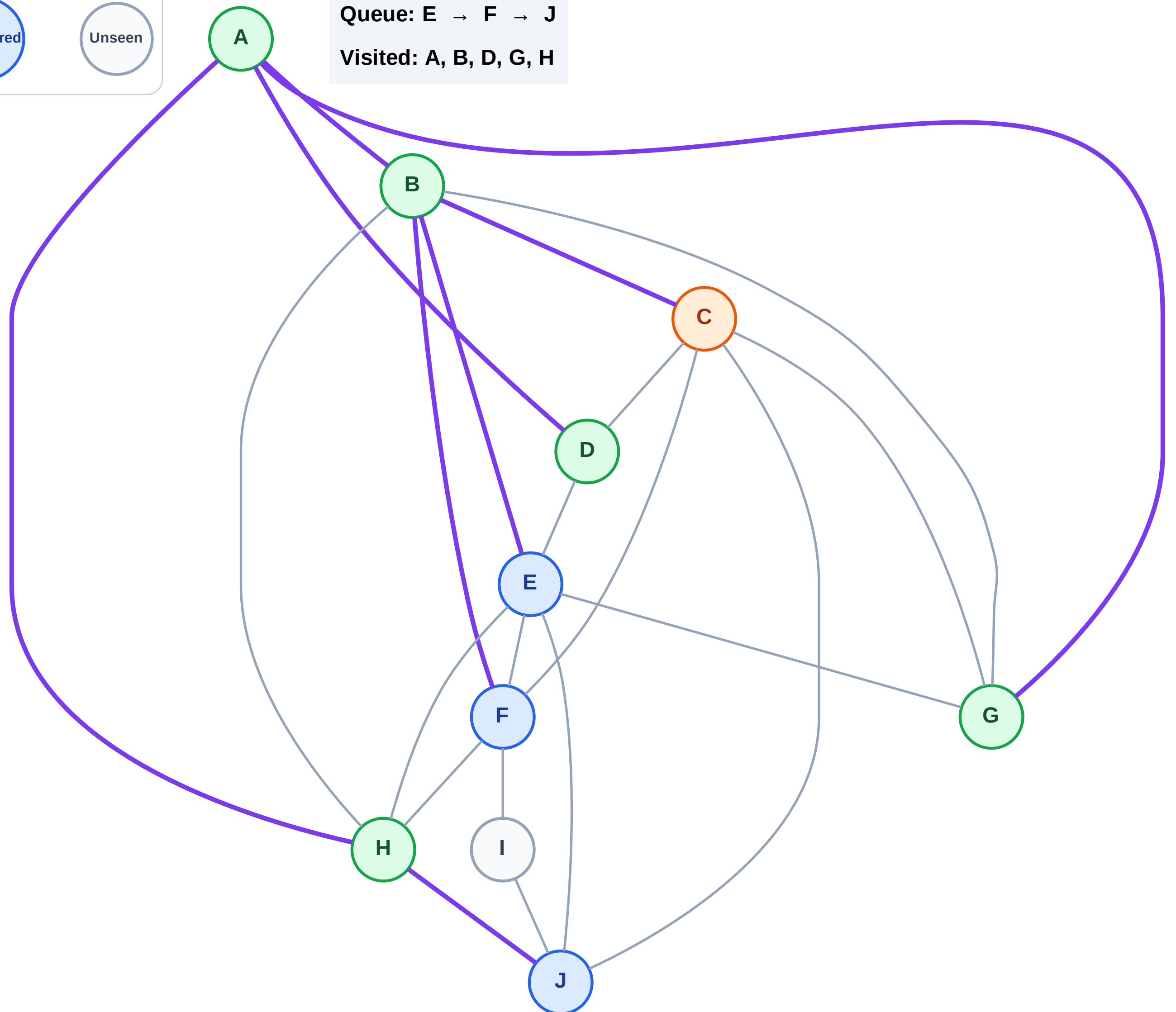


BFS Traversal

Step 12: Process node C



Queue: E → F → J
Visited: A, B, D, G, H



BFS Traversal

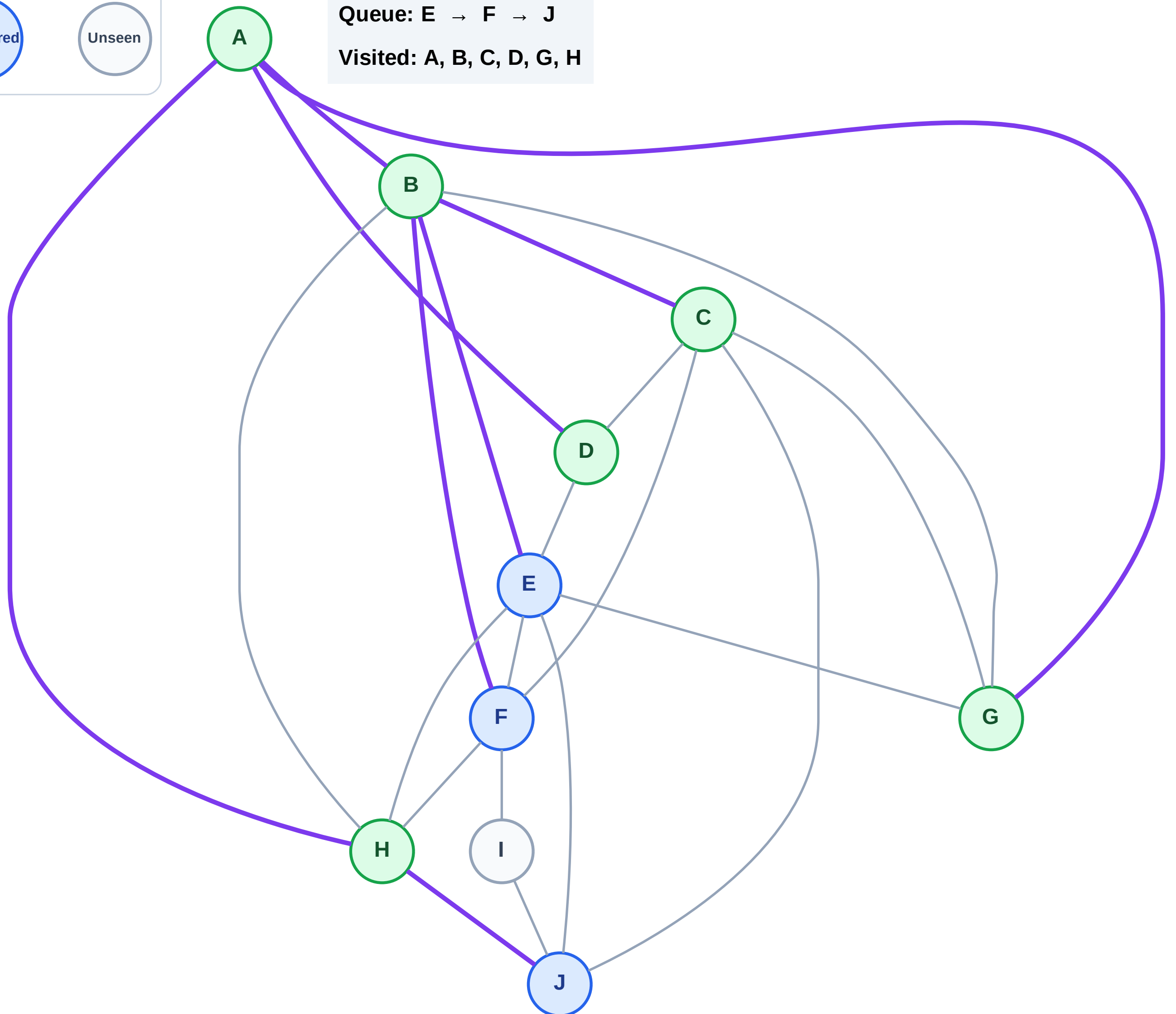
Step 13: No new neighbors from C

Legend



Queue: E → F → J

Visited: A, B, C, D, G, H



BFS Traversal

Step 14: Process node E

Legend

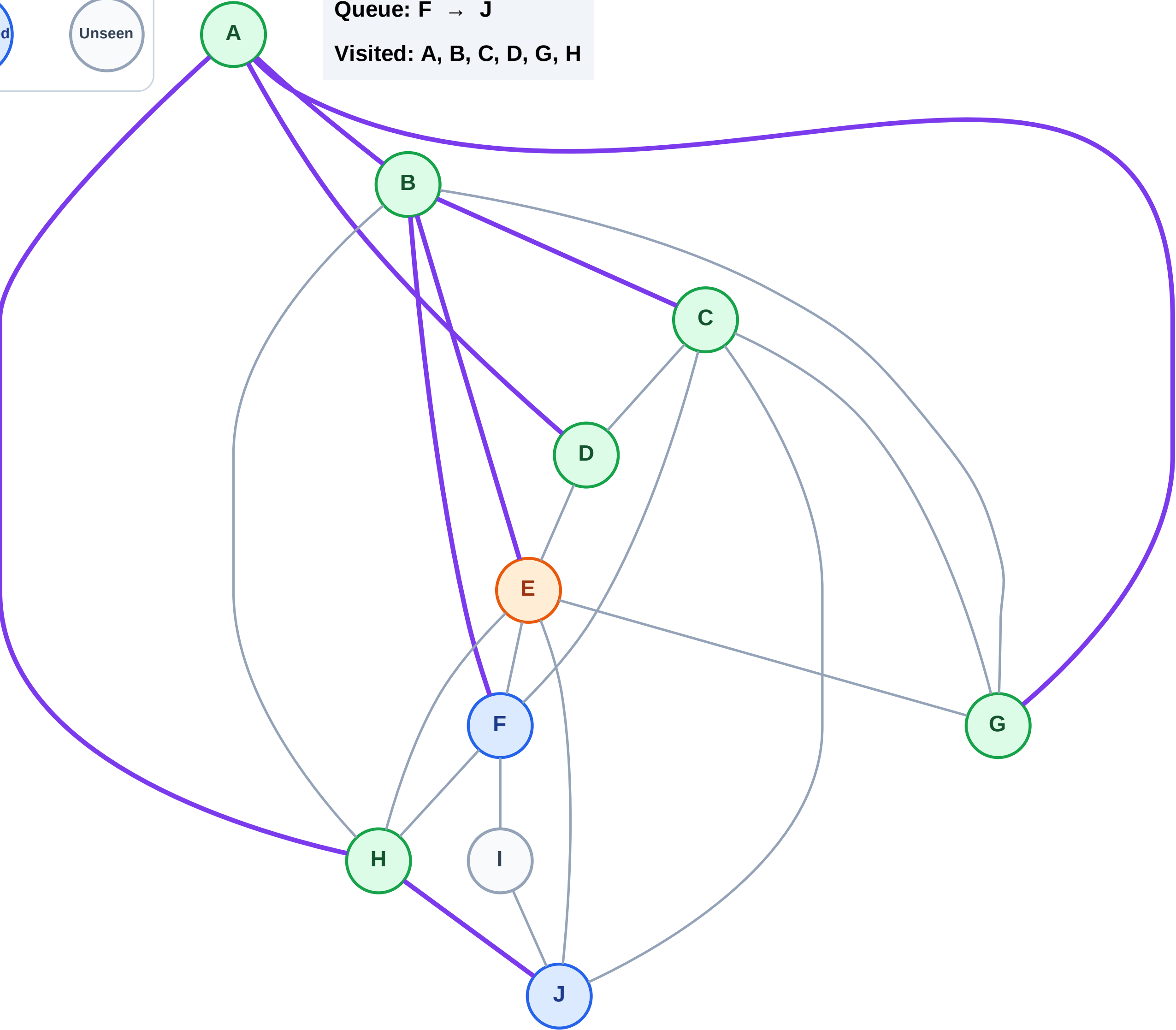
Complete

Processing

Discovered

Unseen

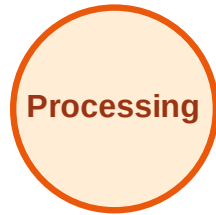
Queue: F → J
Visited: A, B, C, D, G, H



BFS Traversal

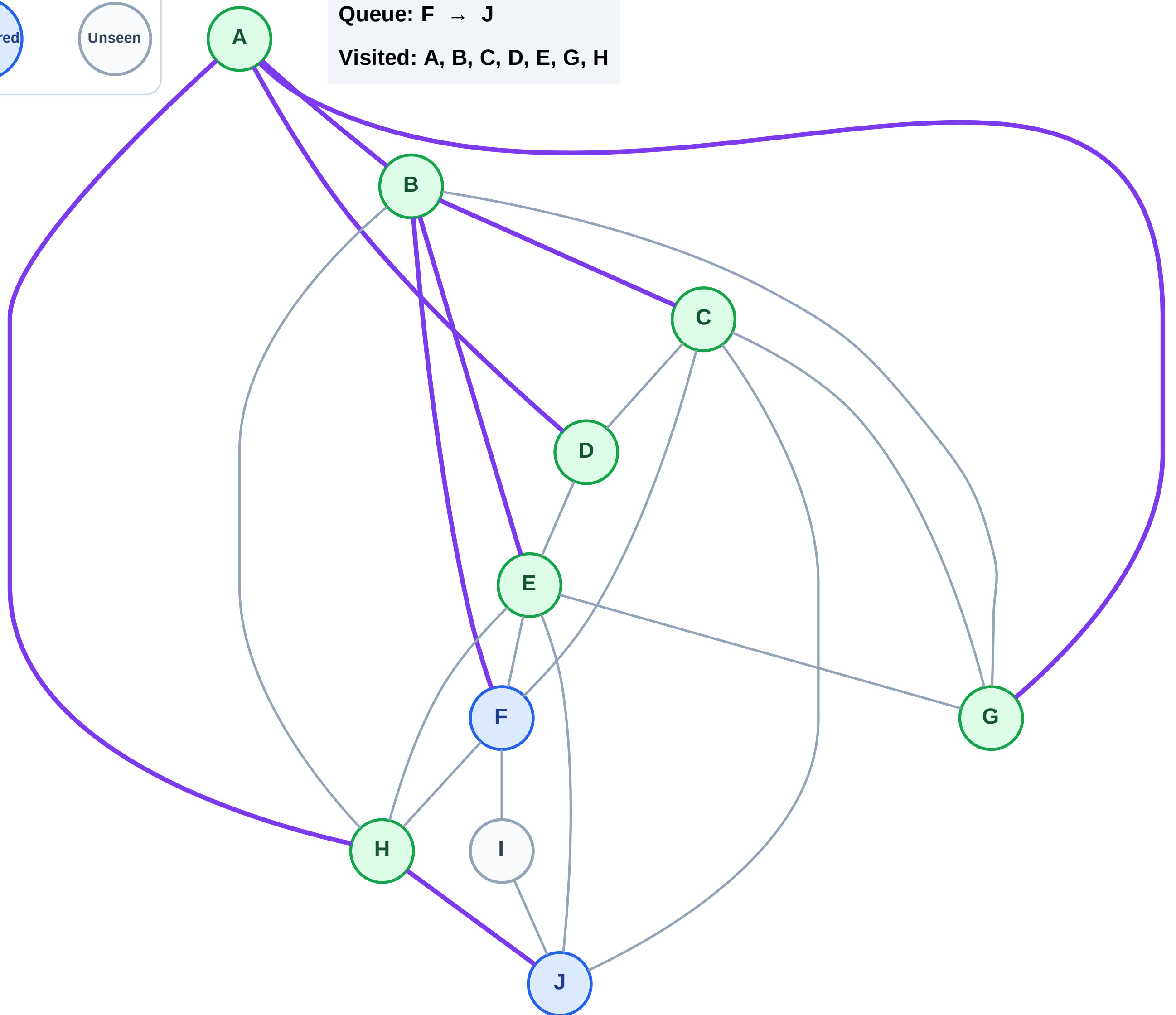
Step 15: No new neighbors from E

Legend



Queue: F → J

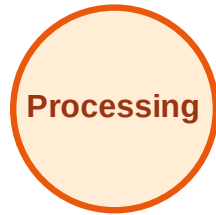
Visited: A, B, C, D, E, G, H



BFS Traversal

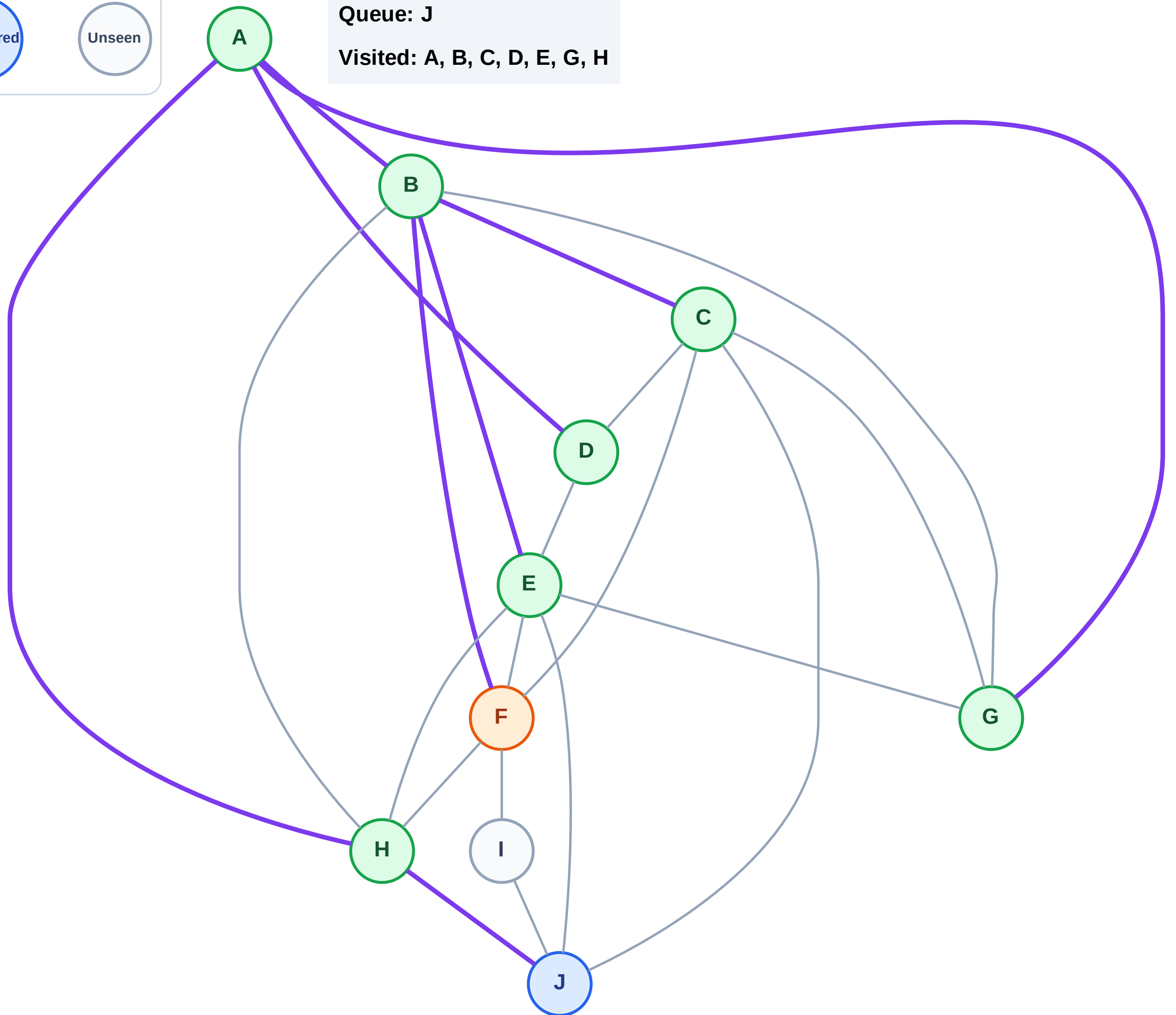
Step 16: Process node F

Legend



Queue: J

Visited: A, B, C, D, E, G, H



BFS Traversal

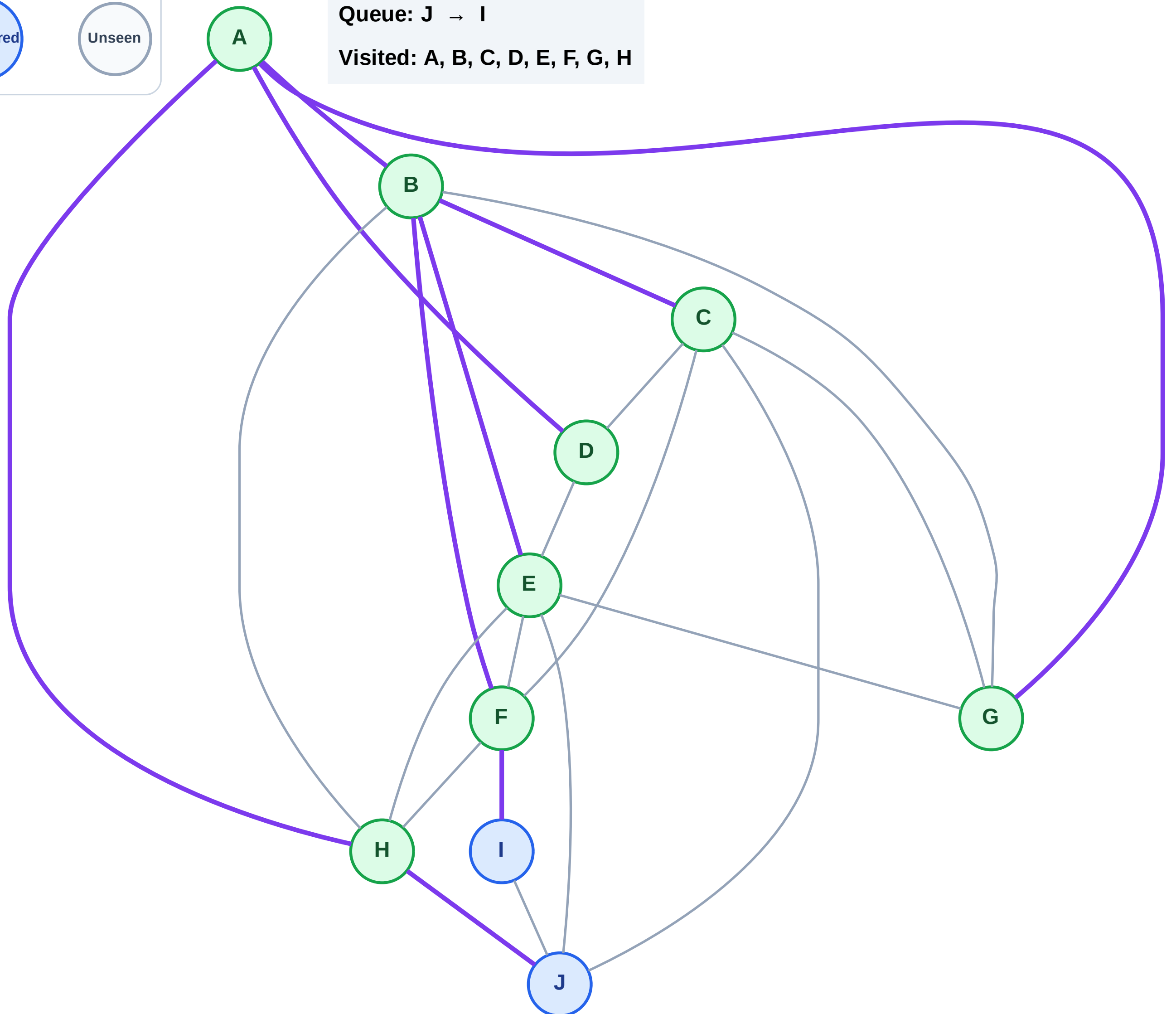
Step 17: Discover I from F

Legend



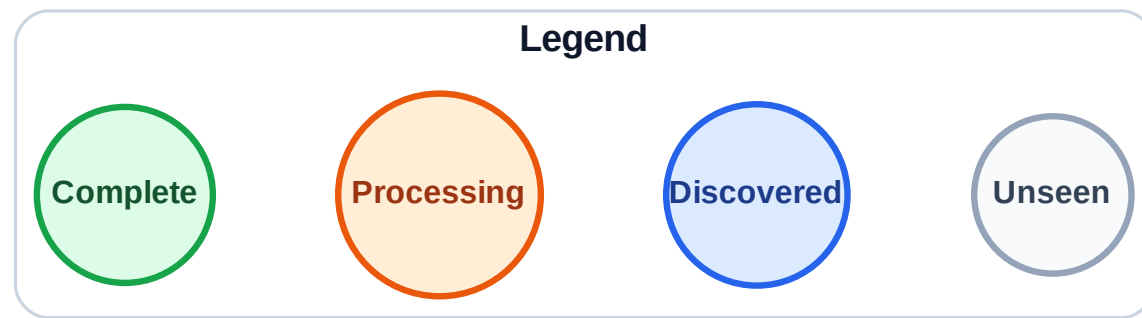
Queue: J → I

Visited: A, B, C, D, E, F, G, H

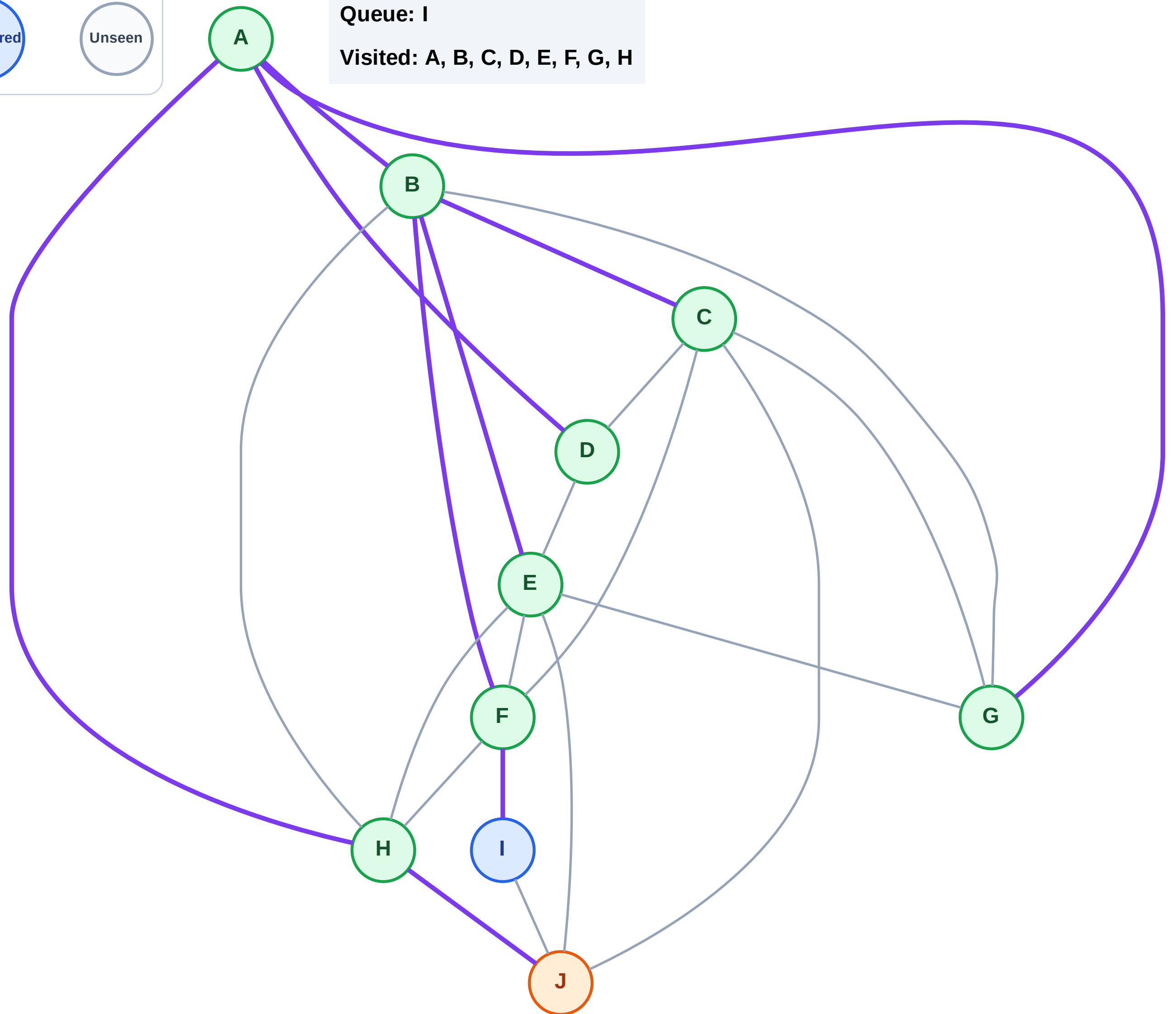


BFS Traversal

Step 18: Process node J



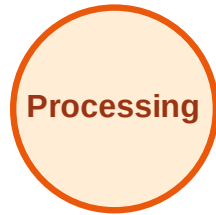
Queue: I
Visited: A, B, C, D, E, F, G, H



BFS Traversal

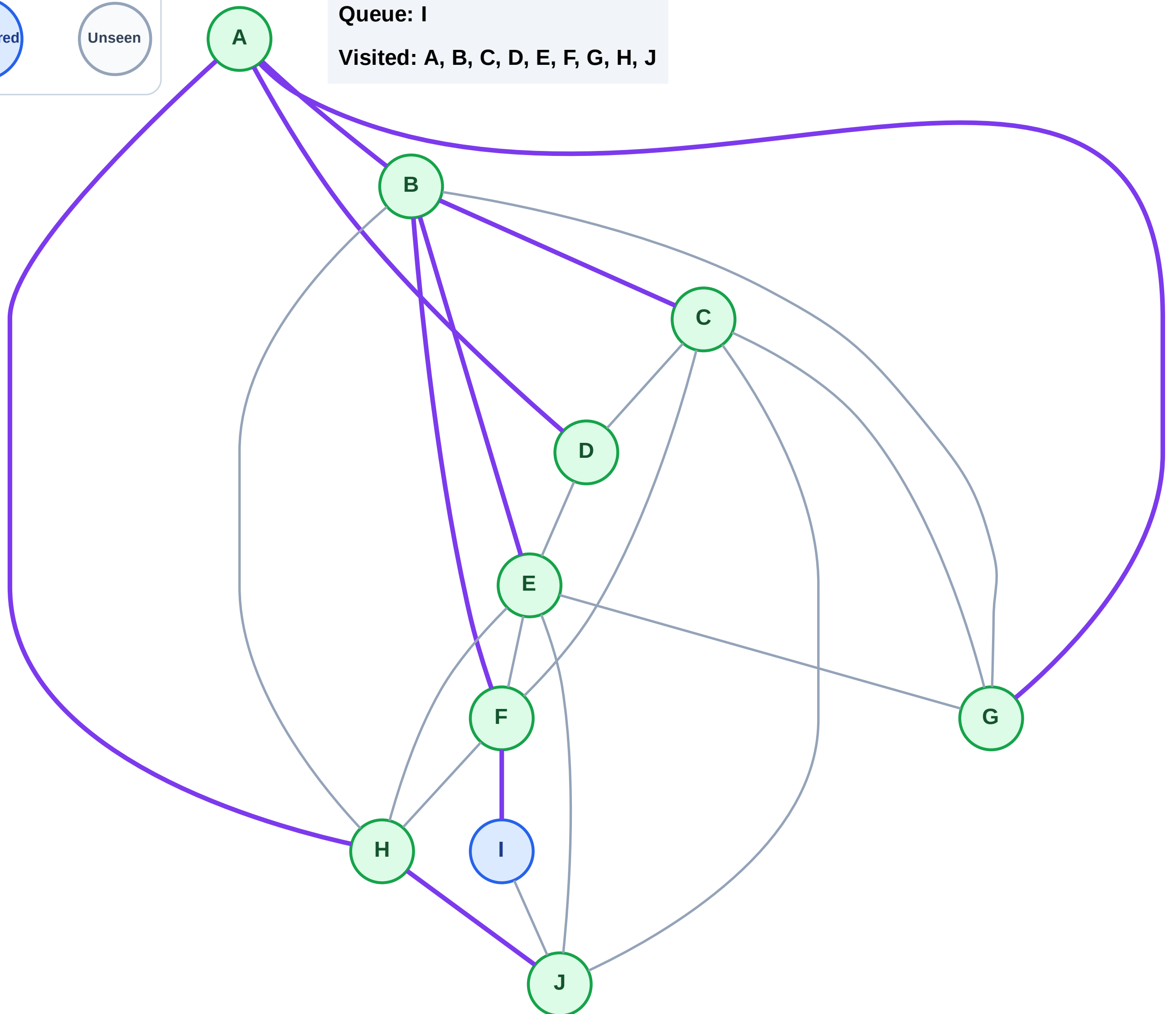
Step 19: No new neighbors from J

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

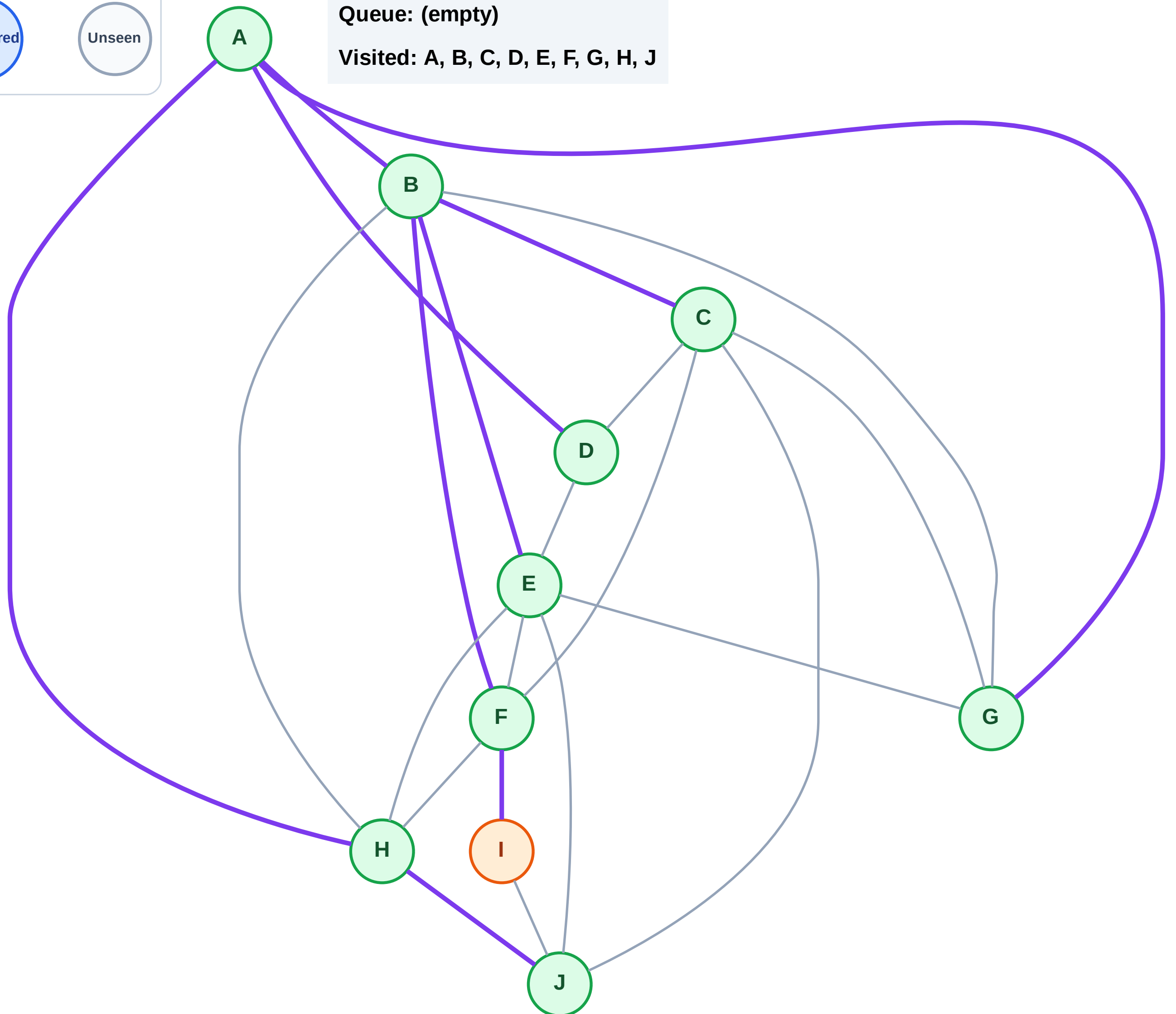
Step 20: Process node I

Legend



Queue: (empty)

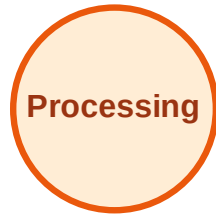
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

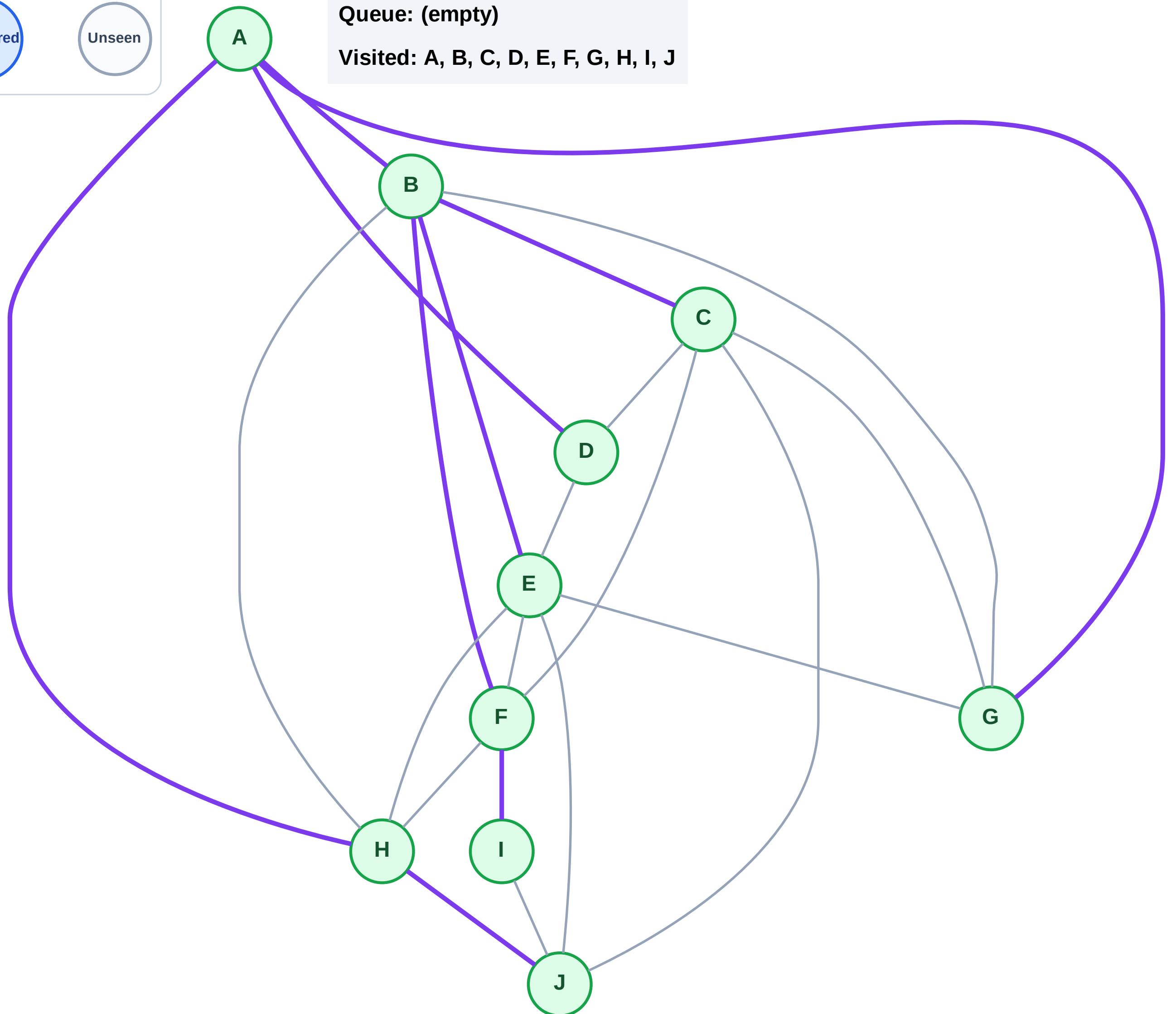
Step 21: No new neighbors from I

Legend



Queue: (empty)

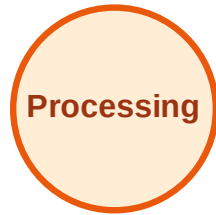
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

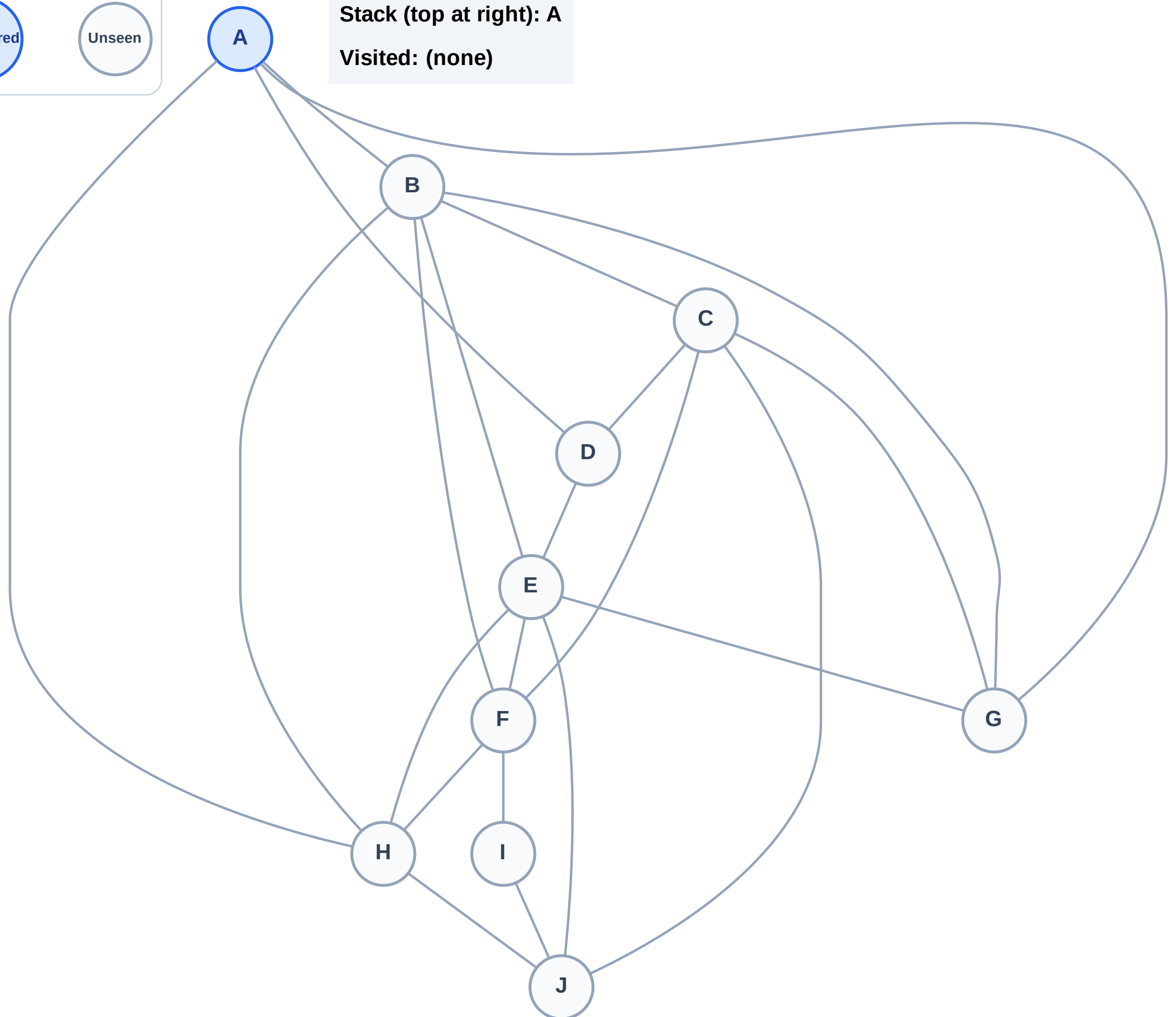
Step 1: Start at A

Legend



Stack (top at right): A

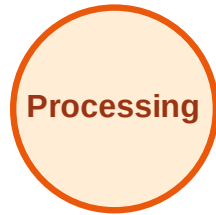
Visited: (none)



DFS Traversal

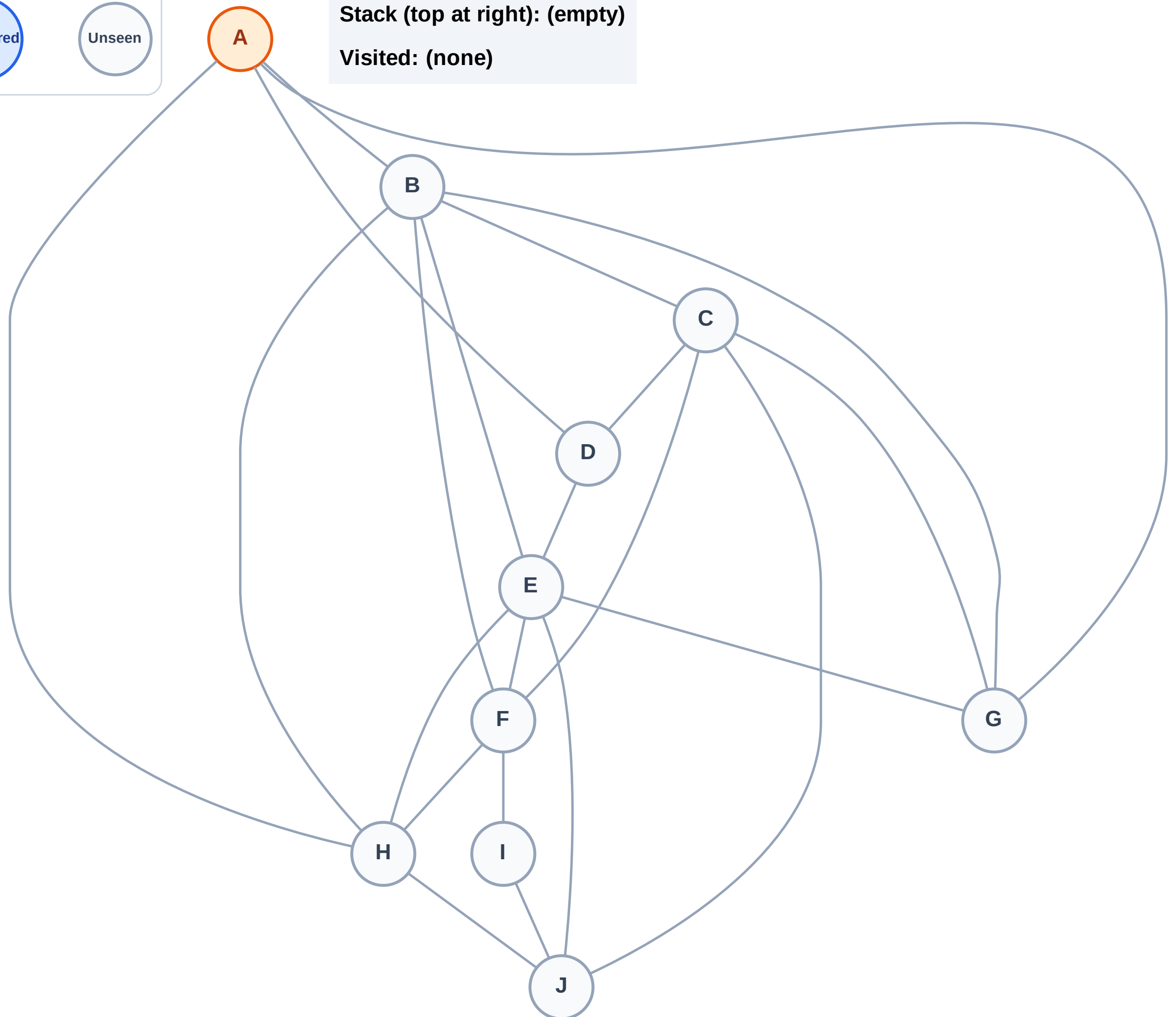
Step 2: Process node A

Legend



Stack (top at right): (empty)

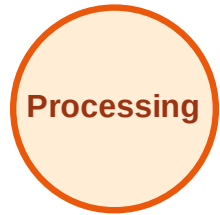
Visited: (none)



DFS Traversal

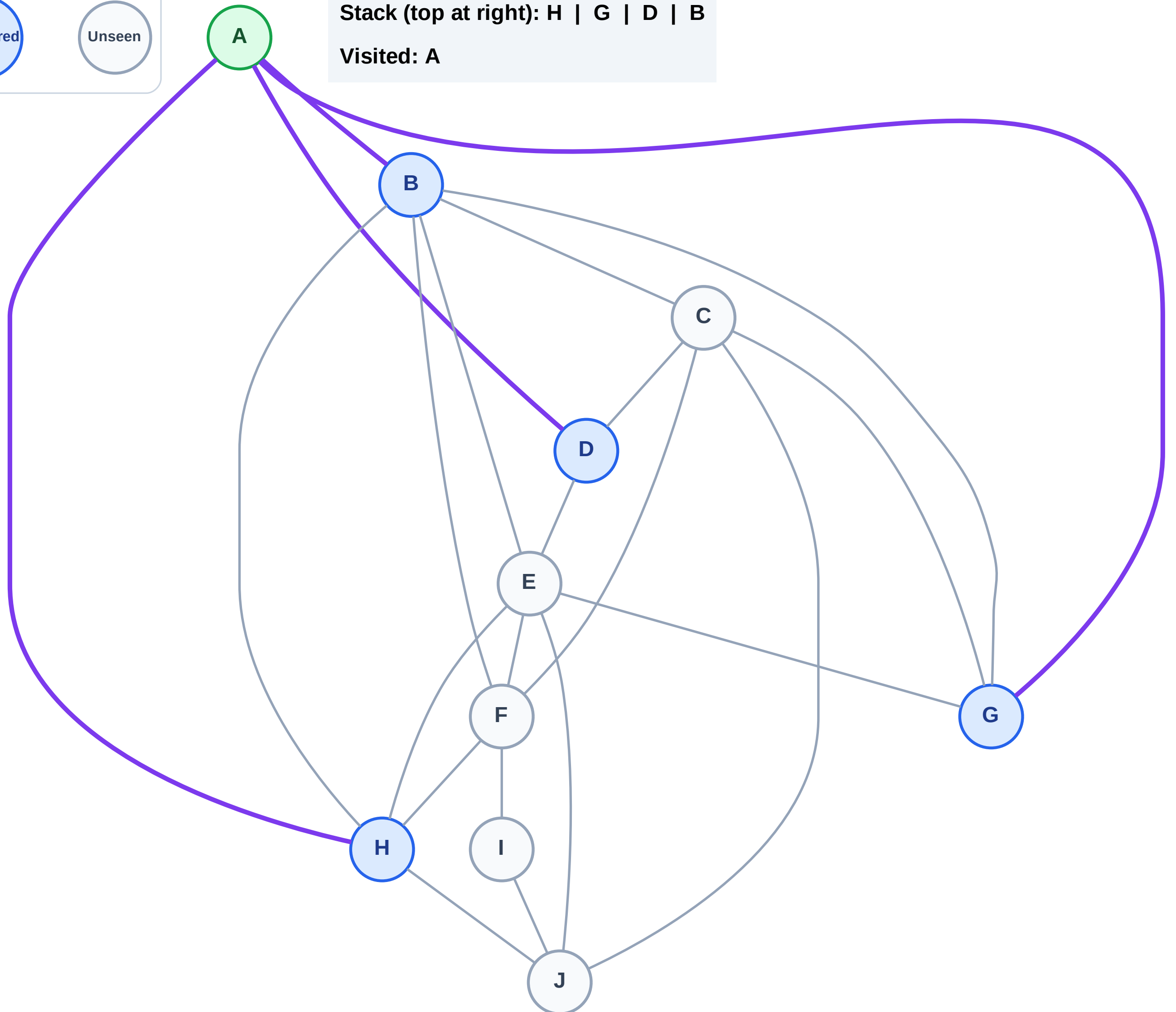
Step 3: Discover H, G, D, B from A

Legend



Stack (top at right): H | G | D | B

Visited: A



Step 4: Process node B

Step 4: Process node B

Legend

Complete

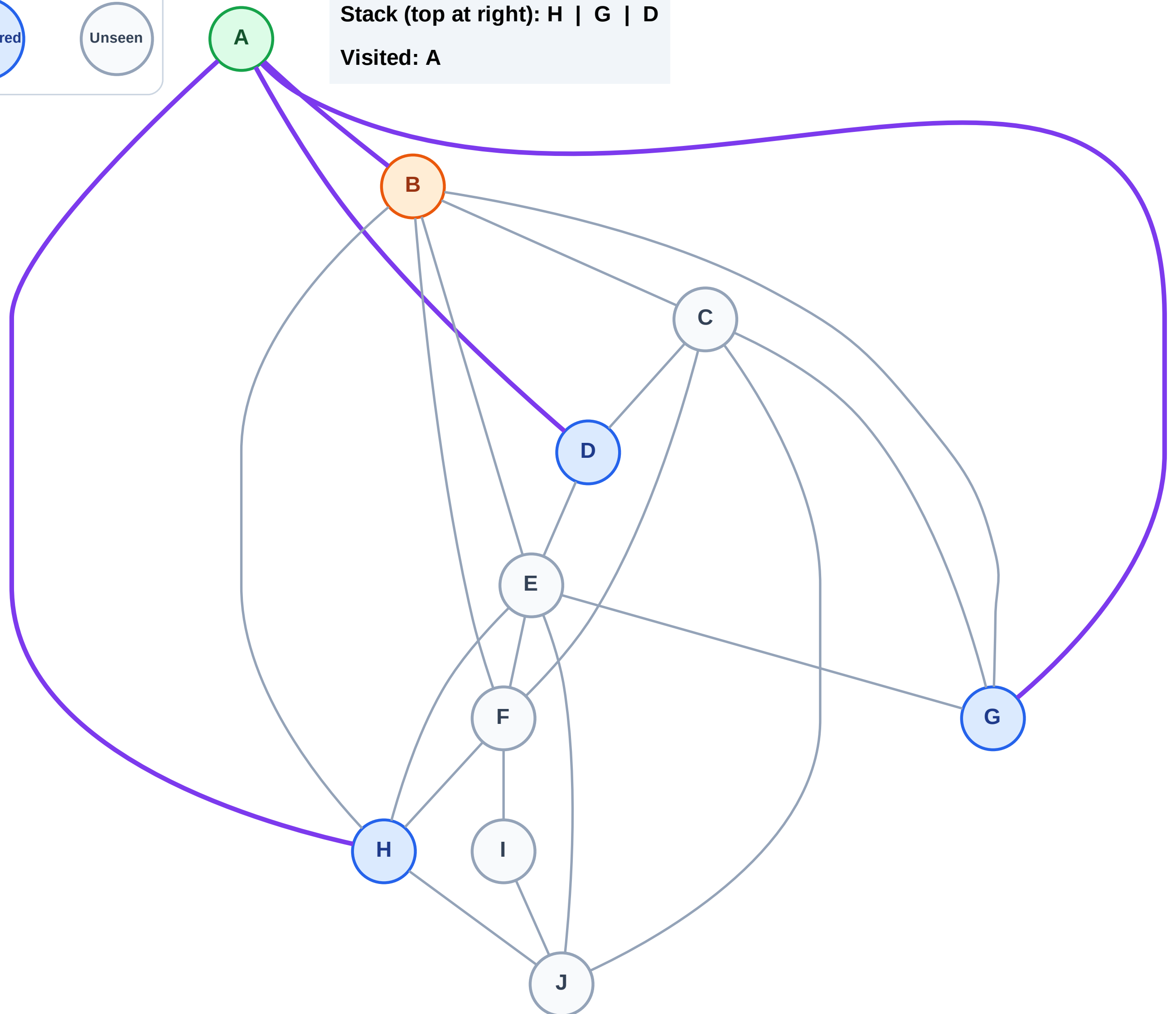
Processing

Discovered

Unseen

Stack (top at right): H | G | D

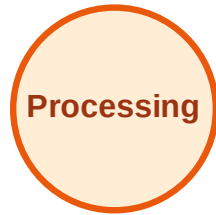
Visited: A



DFS Traversal

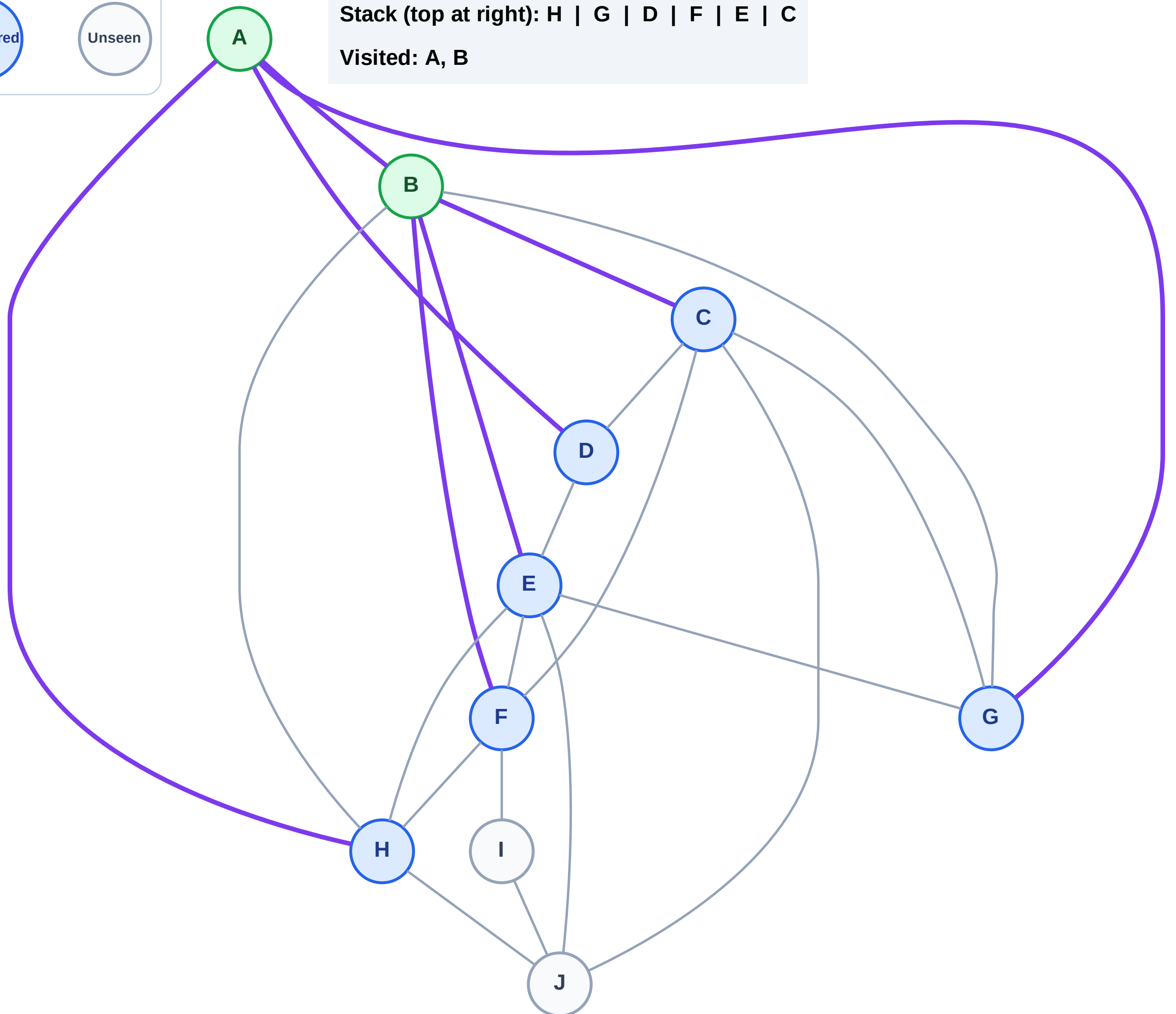
Step 5: Discover F, E, C from B

Legend



Stack (top at right): H | G | D | F | E | C

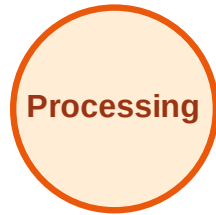
Visited: A, B



DFS Traversal

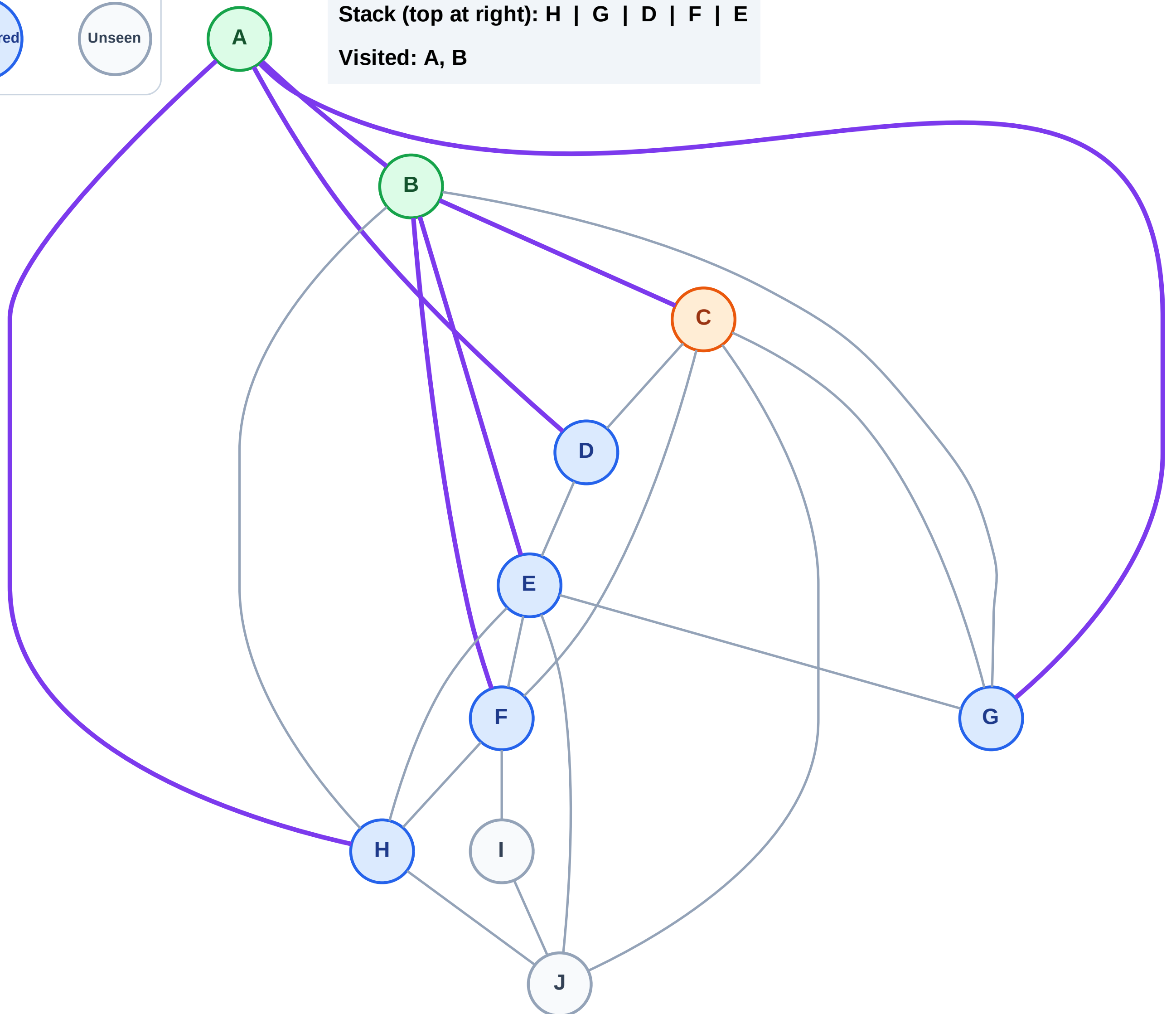
Step 6: Process node C

Legend



Stack (top at right): H | G | D | F | E

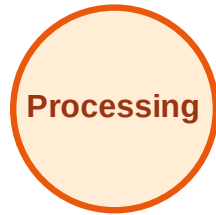
Visited: A, B



DFS Traversal

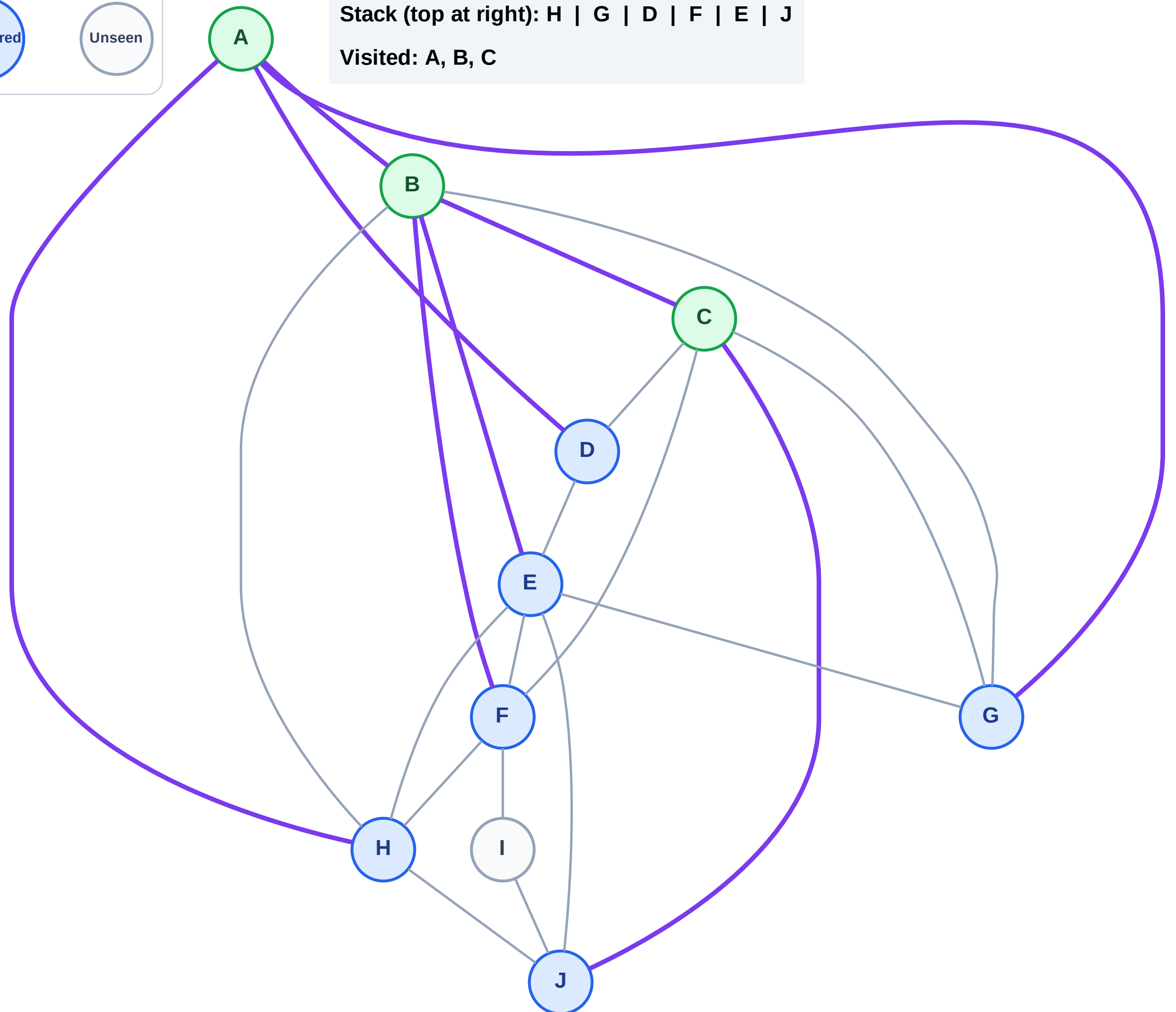
Step 7: Discover J from C

Legend



Stack (top at right): H | G | D | F | E | J

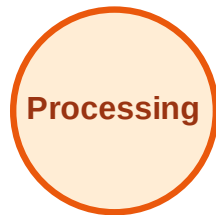
Visited: A, B, C



DFS Traversal

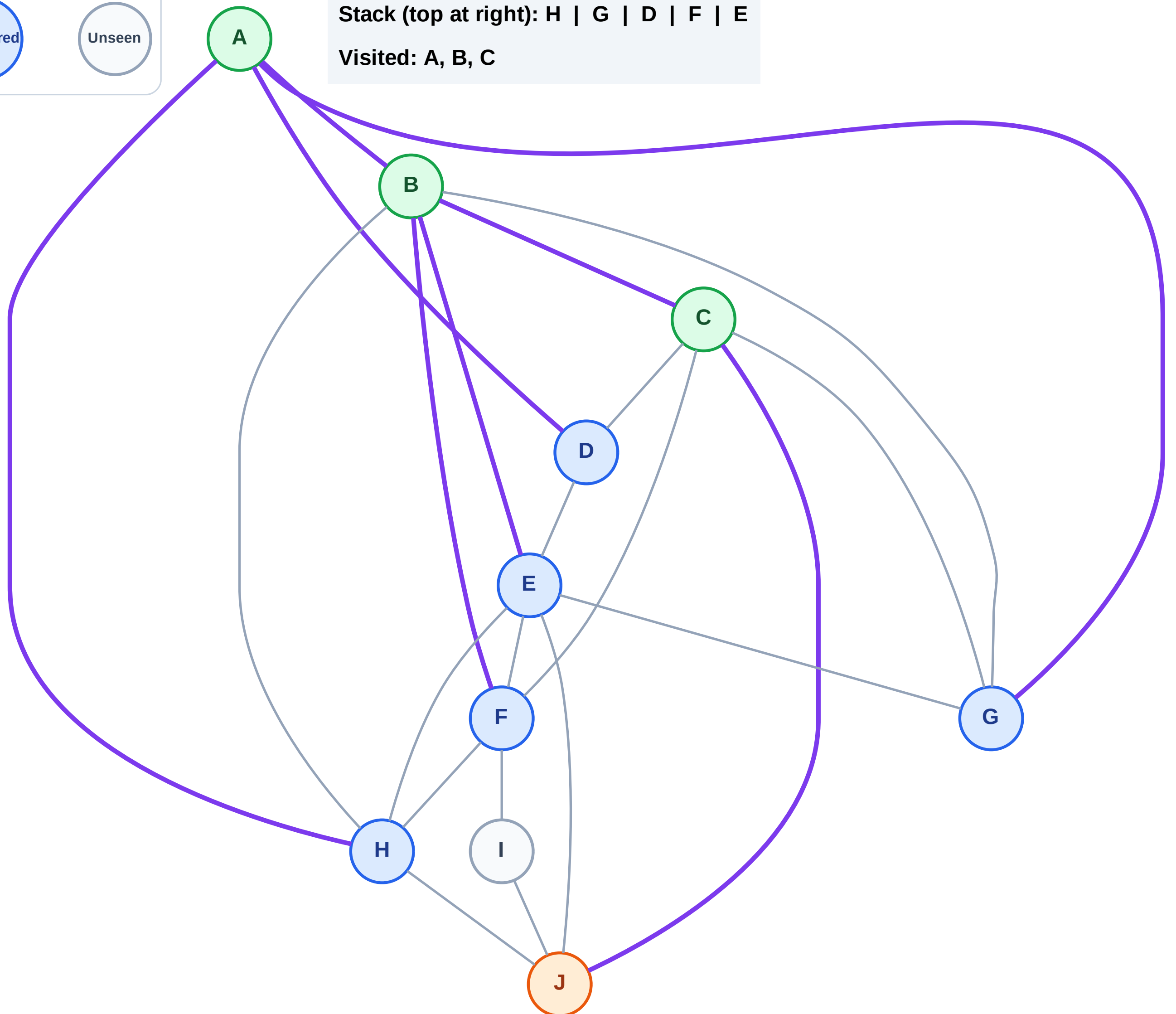
Step 8: Process node J

Legend



Stack (top at right): H | G | D | F | E

Visited: A, B, C



DFS Traversal

Step 9: Discover I from J

Legend

Complete

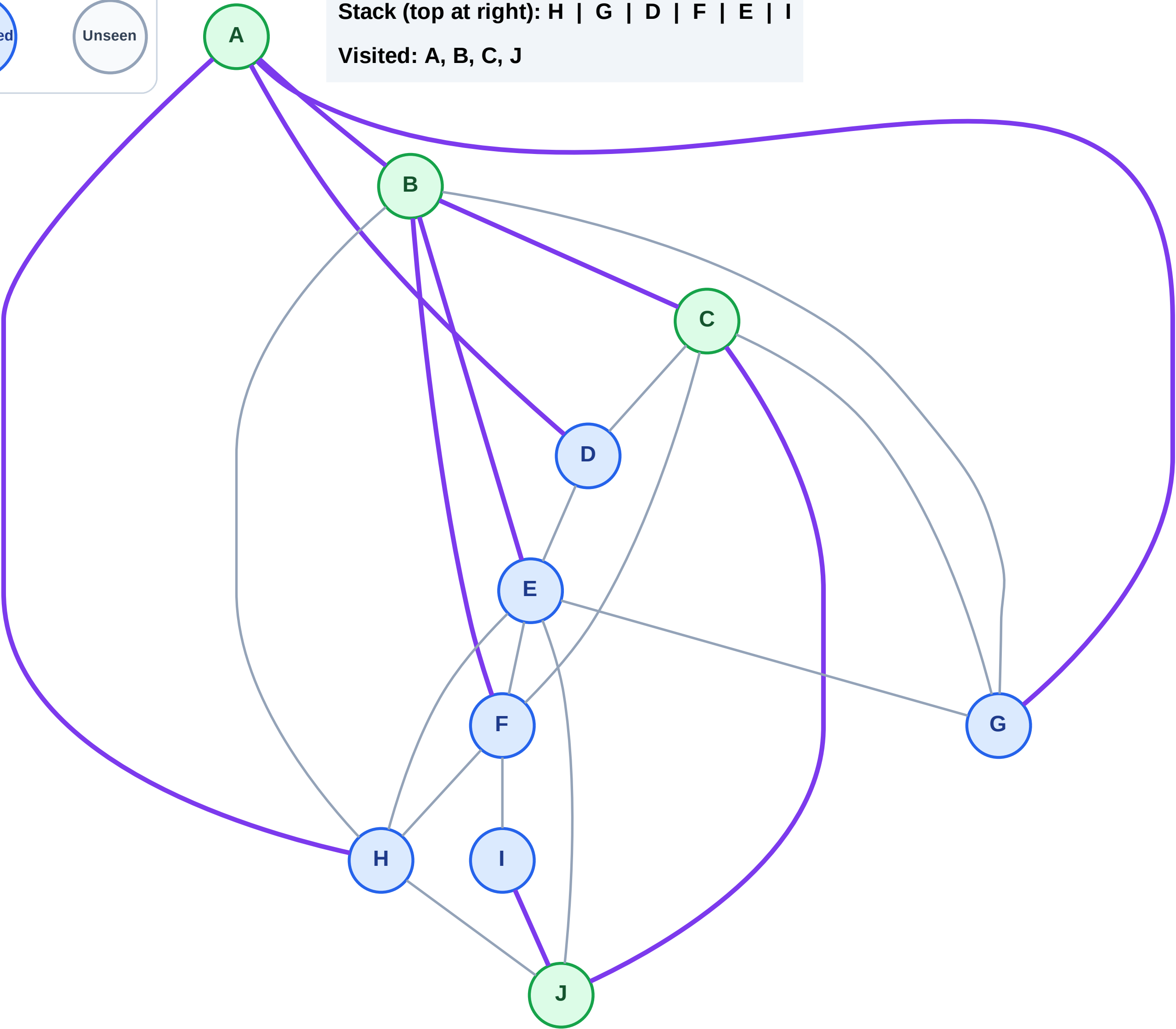
Processing

Discovered

Unseen

Stack (top at right): H | G | D | F | E | I

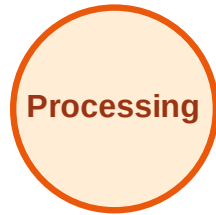
Visited: A, B, C, J



DFS Traversal

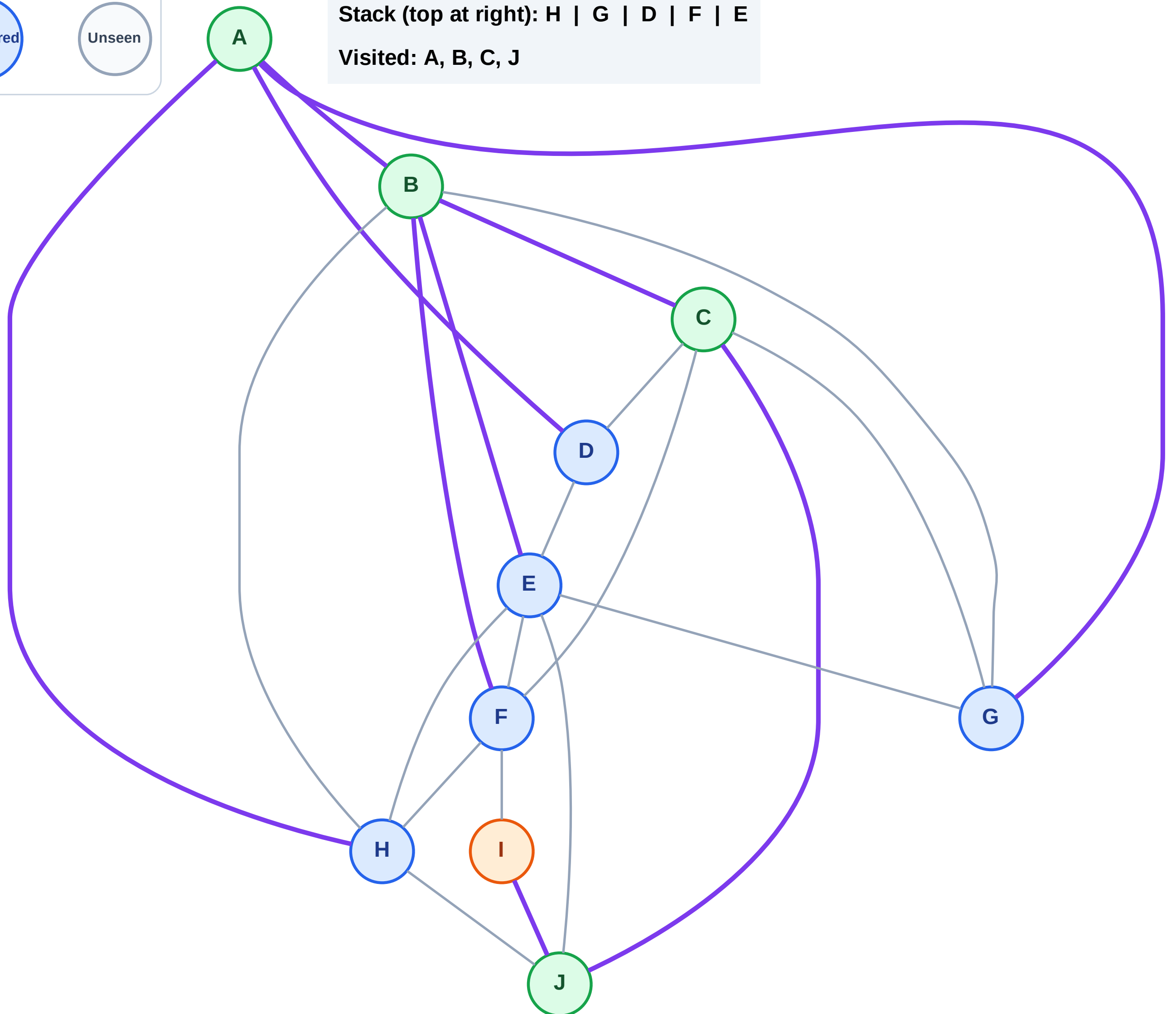
Step 10: Process node I

Legend



Stack (top at right): H | G | D | F | E

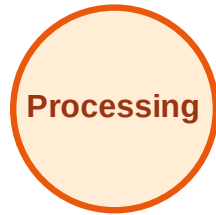
Visited: A, B, C, J



DFS Traversal

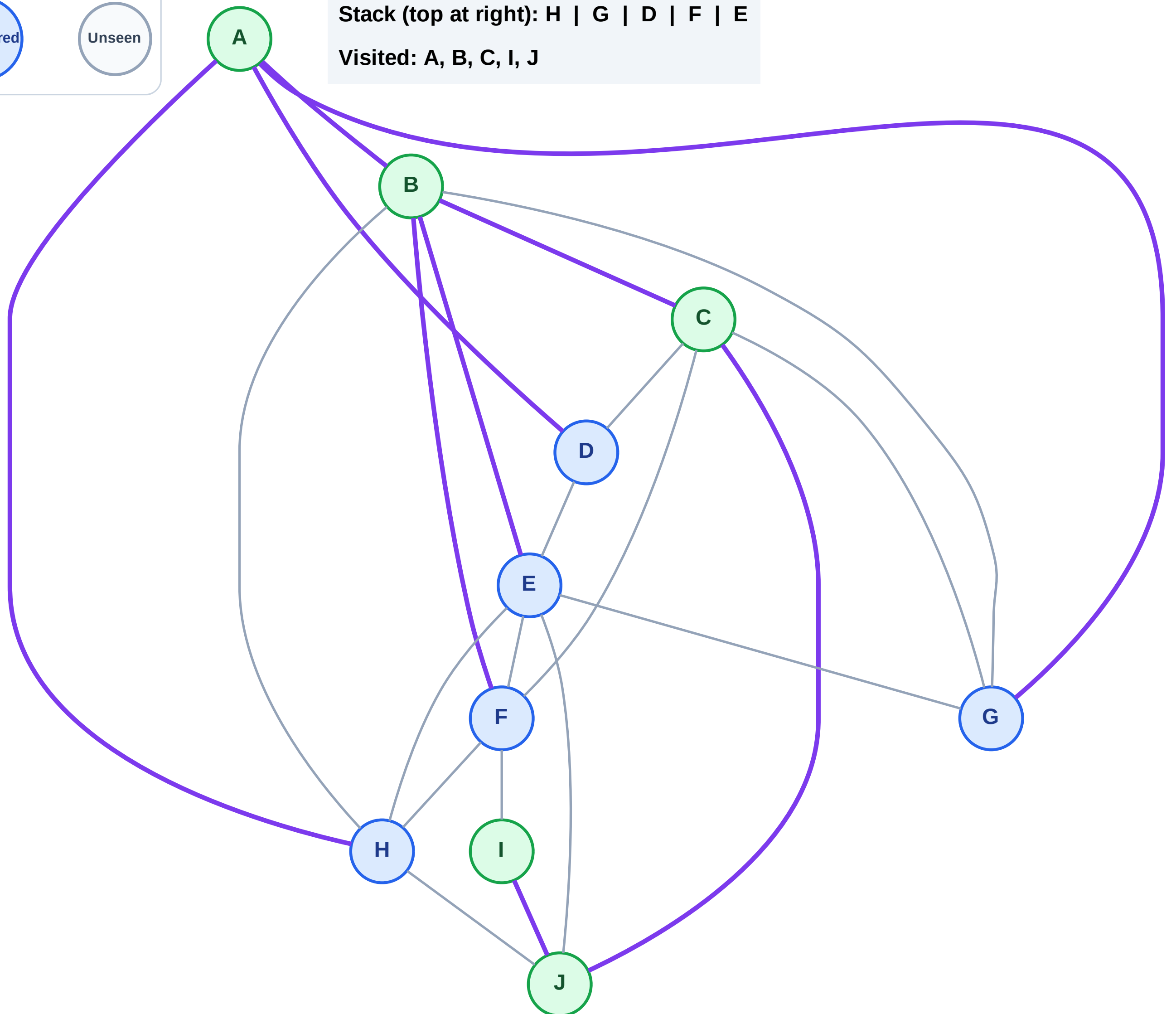
Step 11: No new neighbors from I

Legend



Stack (top at right): H | G | D | F | E

Visited: A, B, C, I, J



DFS Traversal

Step 12: Process node E

Legend

Complete

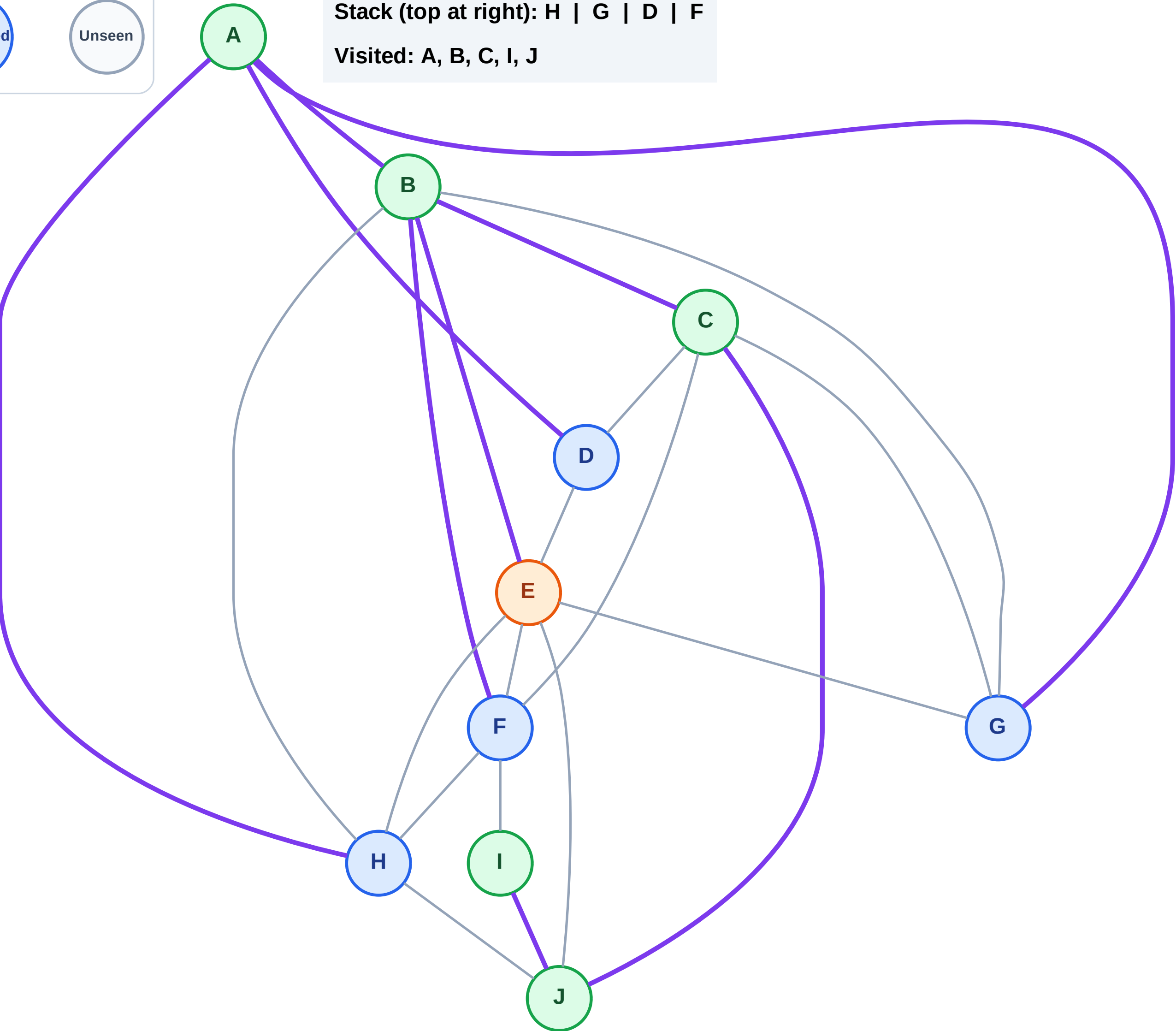
Processing

Discovered

Unseen

Stack (top at right): H | G | D | F

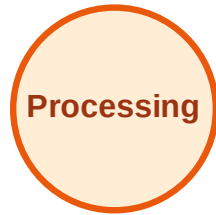
Visited: A, B, C, I, J



DFS Traversal

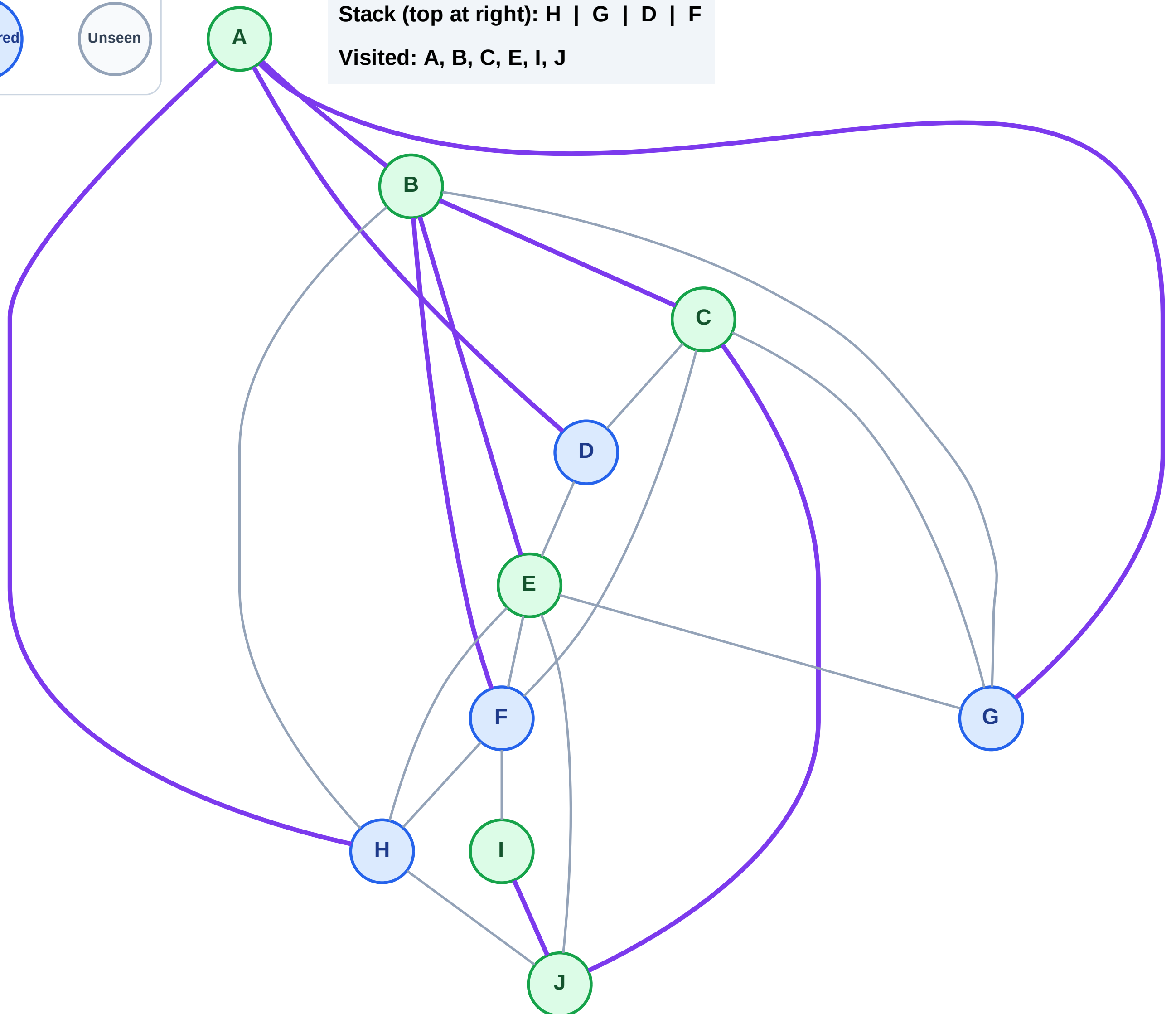
Step 13: No new neighbors from E

Legend



Stack (top at right): H | G | D | F

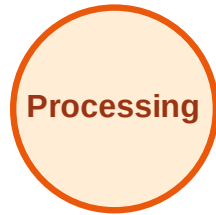
Visited: A, B, C, E, I, J



DFS Traversal

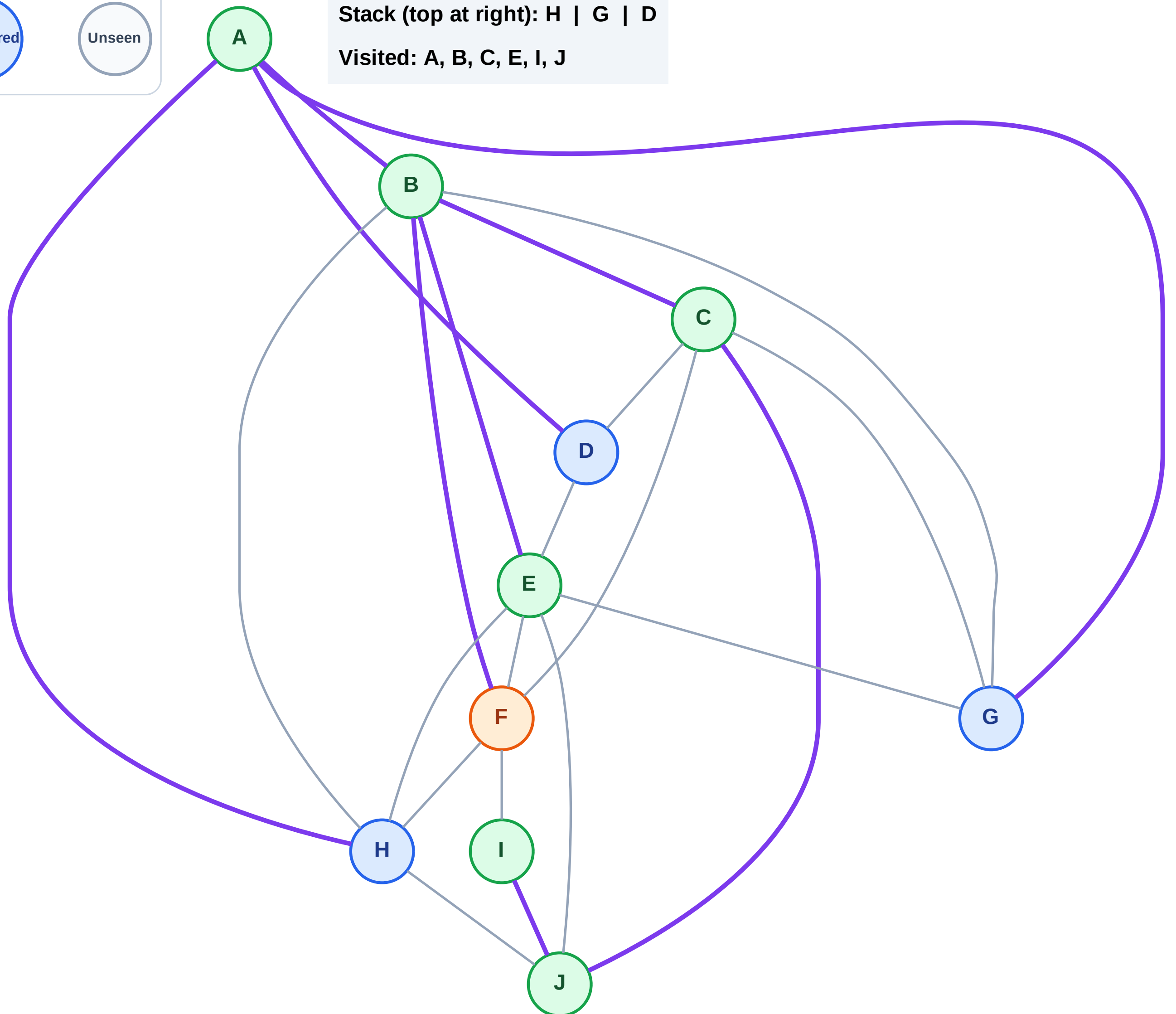
Step 14: Process node F

Legend



Stack (top at right): H | G | D

Visited: A, B, C, E, I, J



DFS Traversal

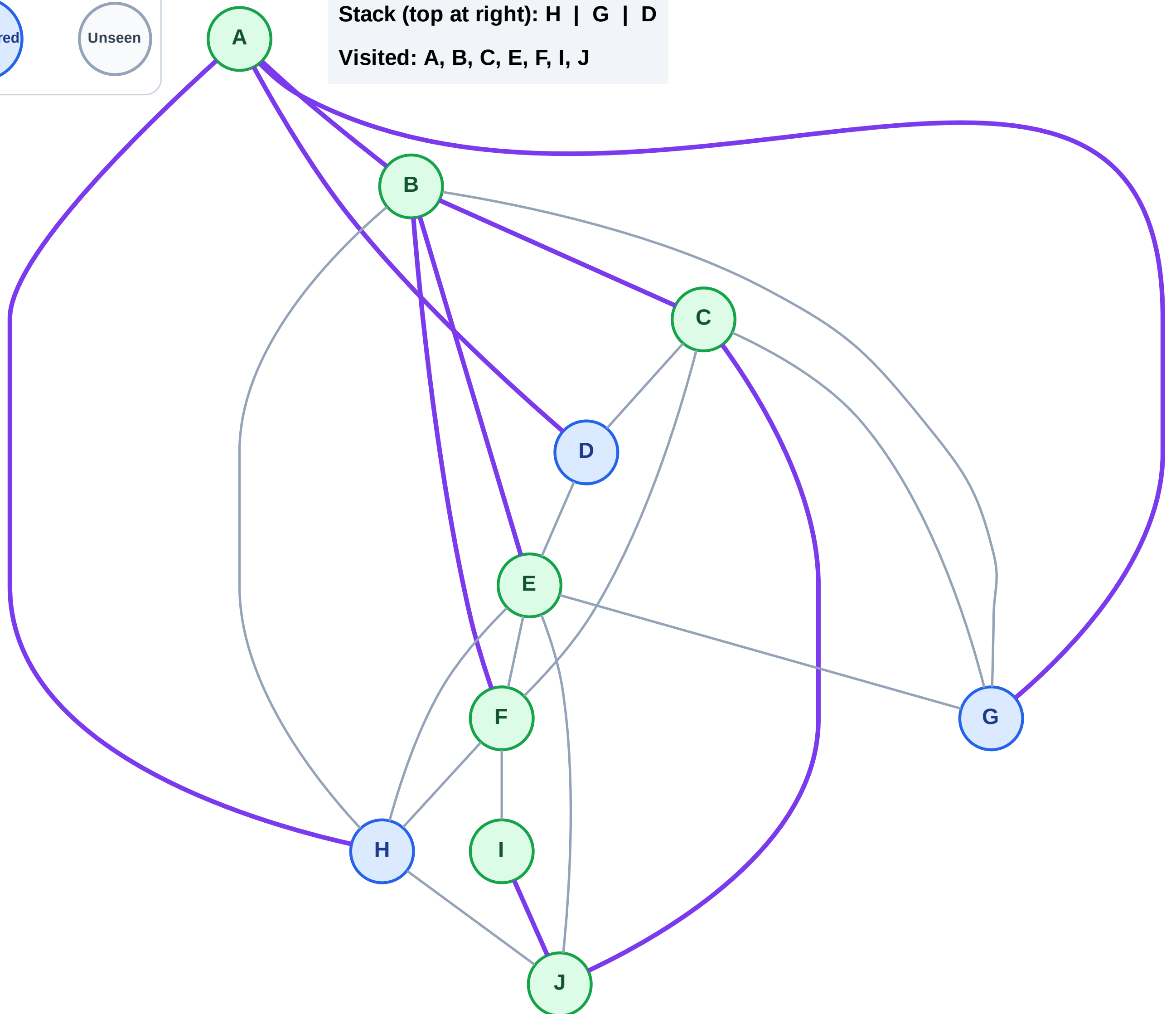
Step 15: No new neighbors from F

Legend



Stack (top at right): H | G | D

Visited: A, B, C, E, F, I, J



DFS Traversal

Step 16: Process node D

Legend

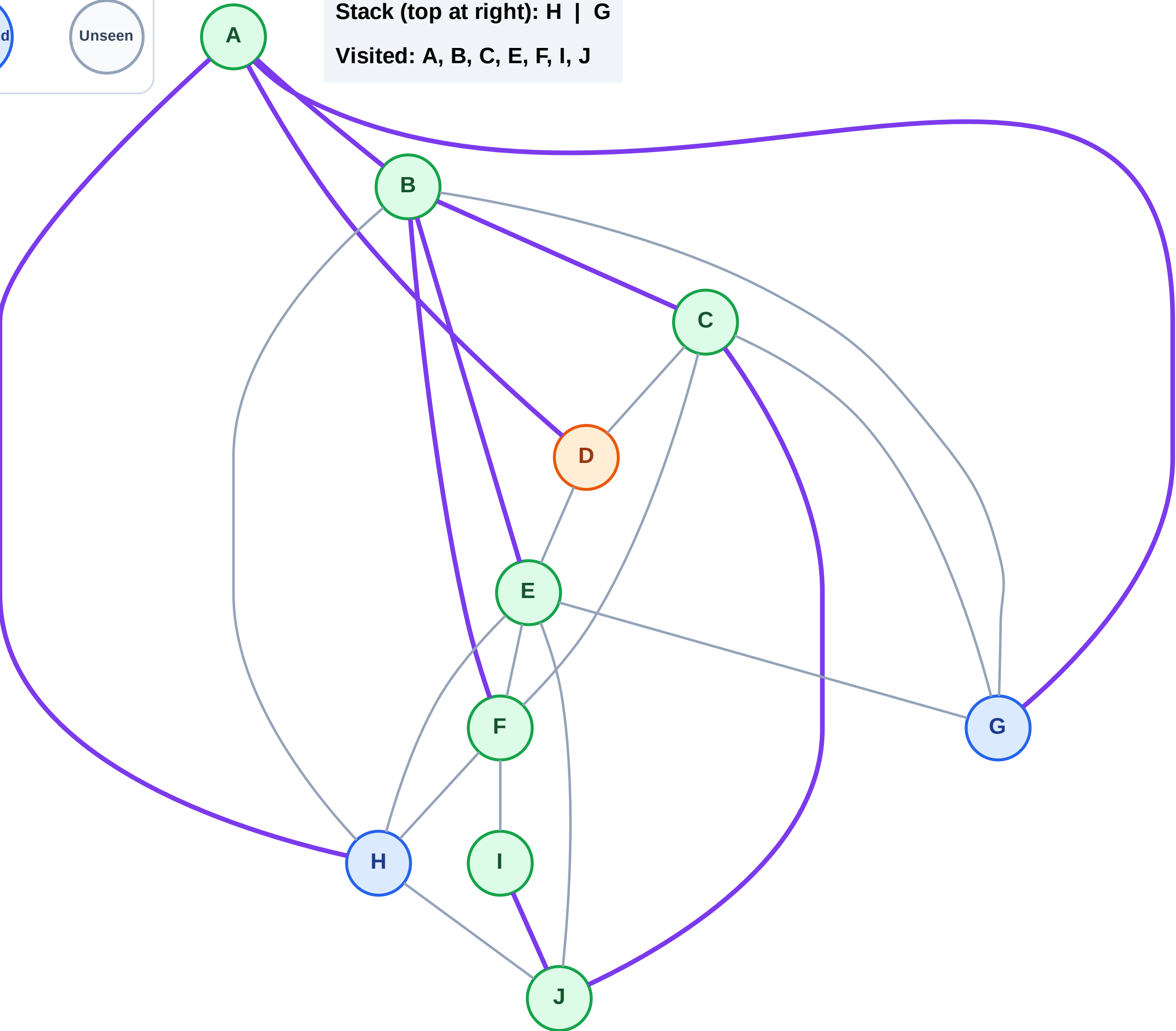
Complete

Processing

Discovered

Unseen

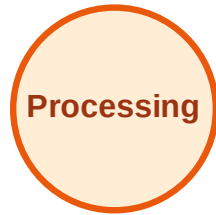
Stack (top at right): H | G
Visited: A, B, C, E, F, I, J



DFS Traversal

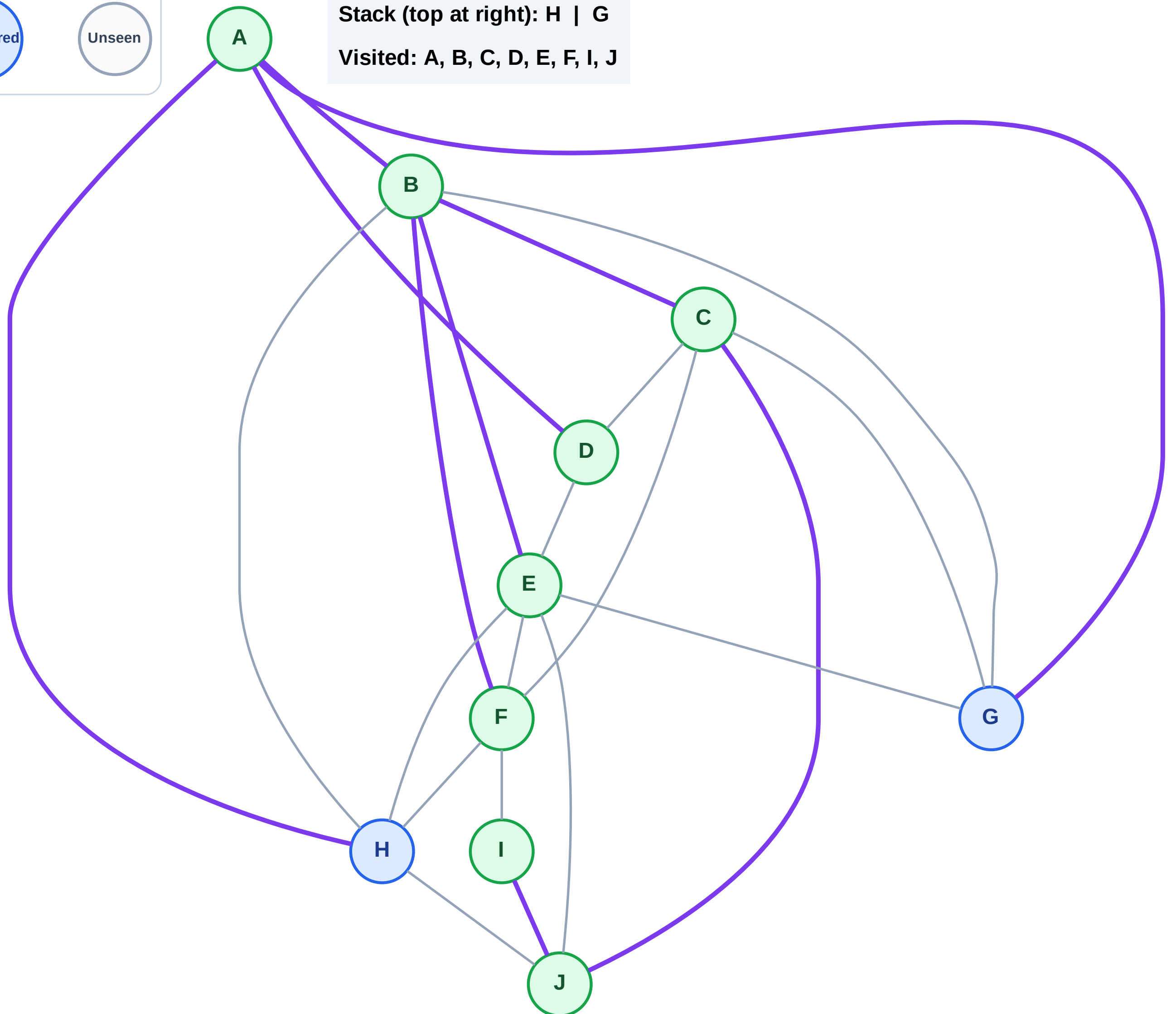
Step 17: No new neighbors from D

Legend



Stack (top at right): H | G

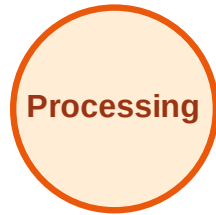
Visited: A, B, C, D, E, F, I, J



DFS Traversal

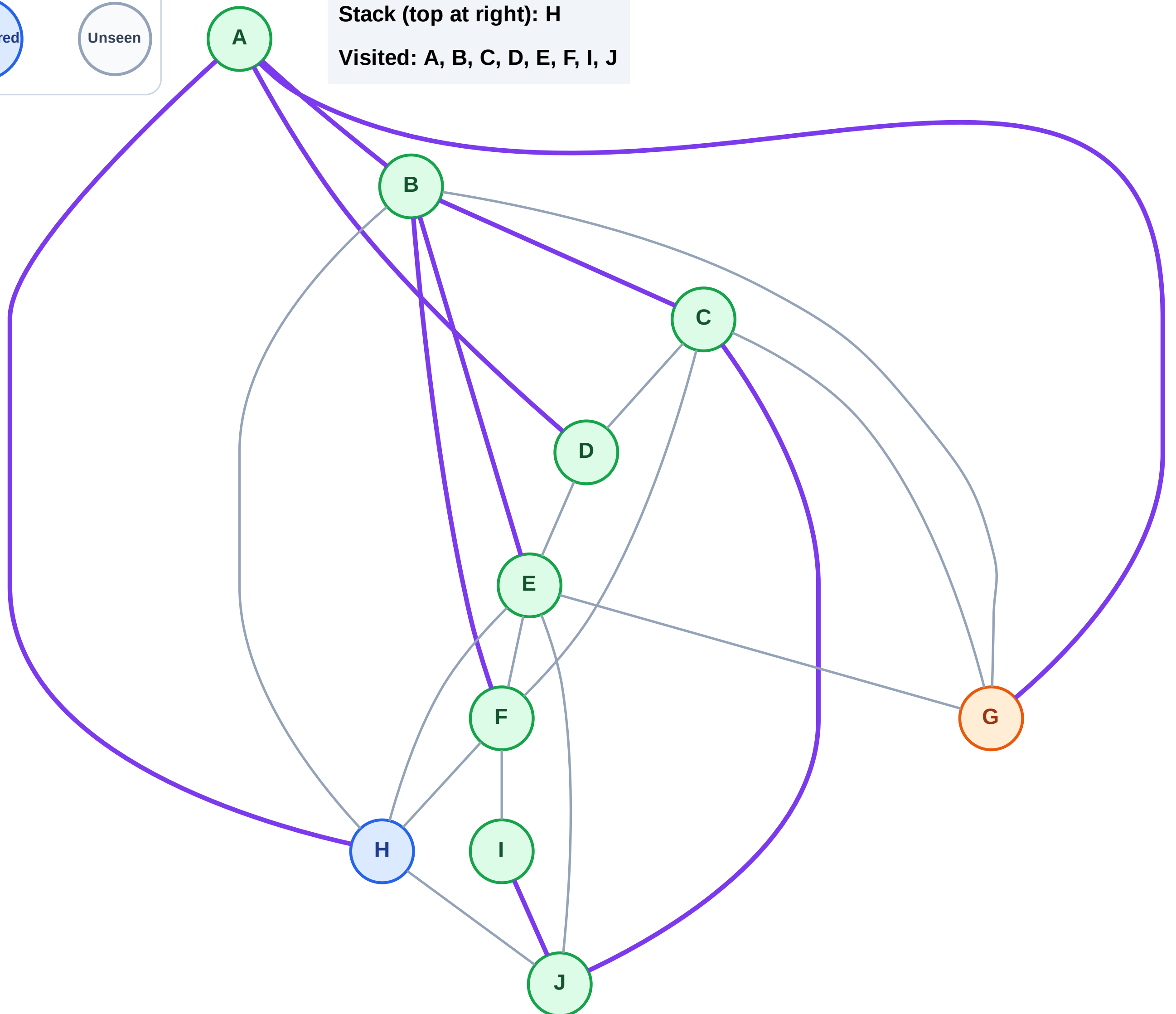
Step 18: Process node G

Legend



Stack (top at right): H

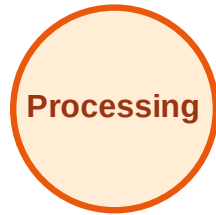
Visited: A, B, C, D, E, F, I, J



DFS Traversal

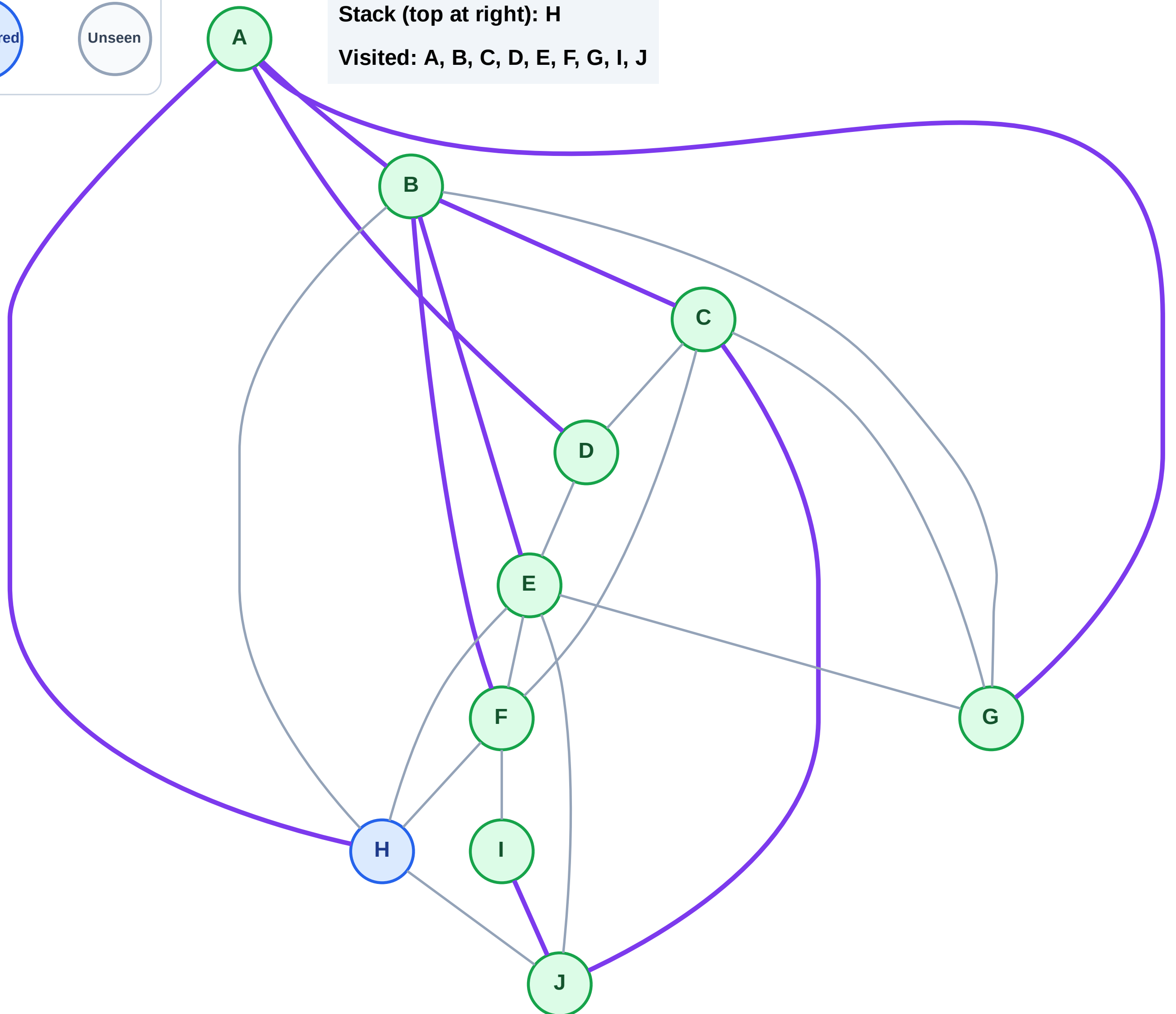
Step 19: No new neighbors from G

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 20: Process node H

Legend

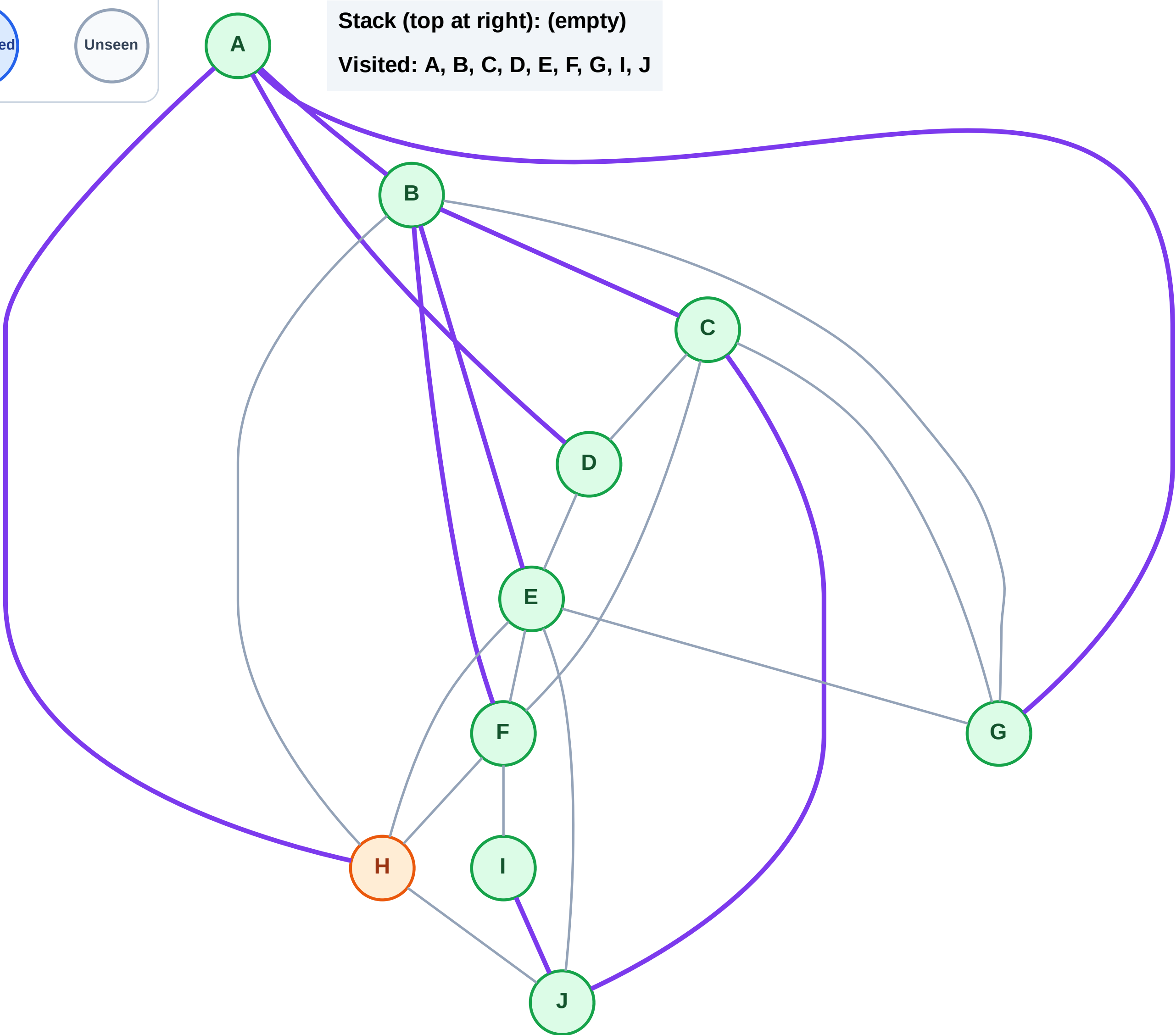
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

